University of DGK
Ghazi University :-

Assignment

Programing Fundamental Lab II.

Subject :-

Programing Fundamental.

Submitted by :-

Rbia Fatima

Submitted To :-

Prof. Asad Majeed.

Department :-

IT Department

BS computer Science

Section E.

Start →

Task-1 :-

Type and Save the following programs in visual Studio - Run these programs and observe their output.

Program 1 :-

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4  char Letter;
5  Letter = 'A';
6  cout<<letter<<'\n';
7  Letter='B';
8  cout <<letter <<'\n';
9  cin >> Letter;
10  return 0; }
```

Output :-

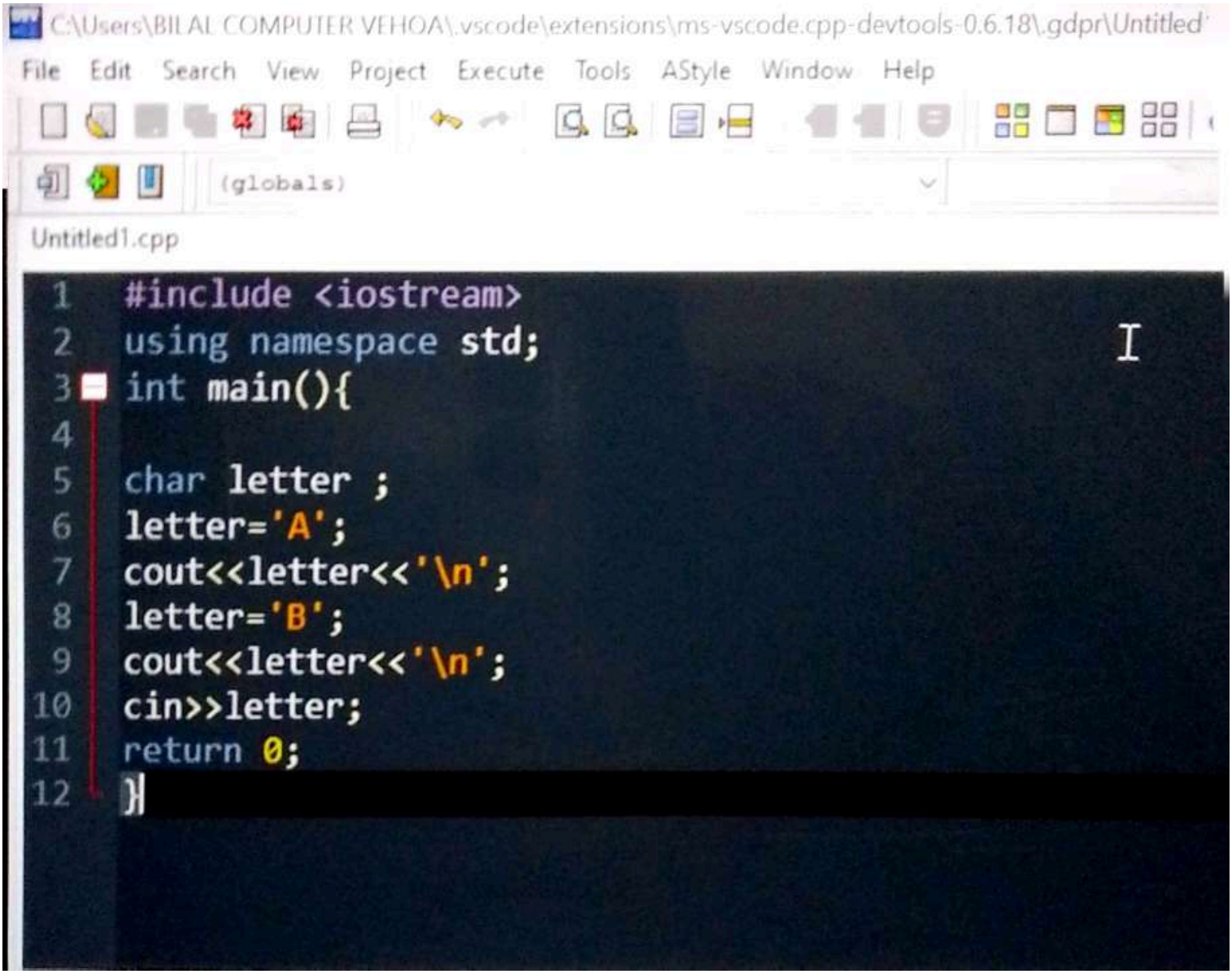
```
A
B
```

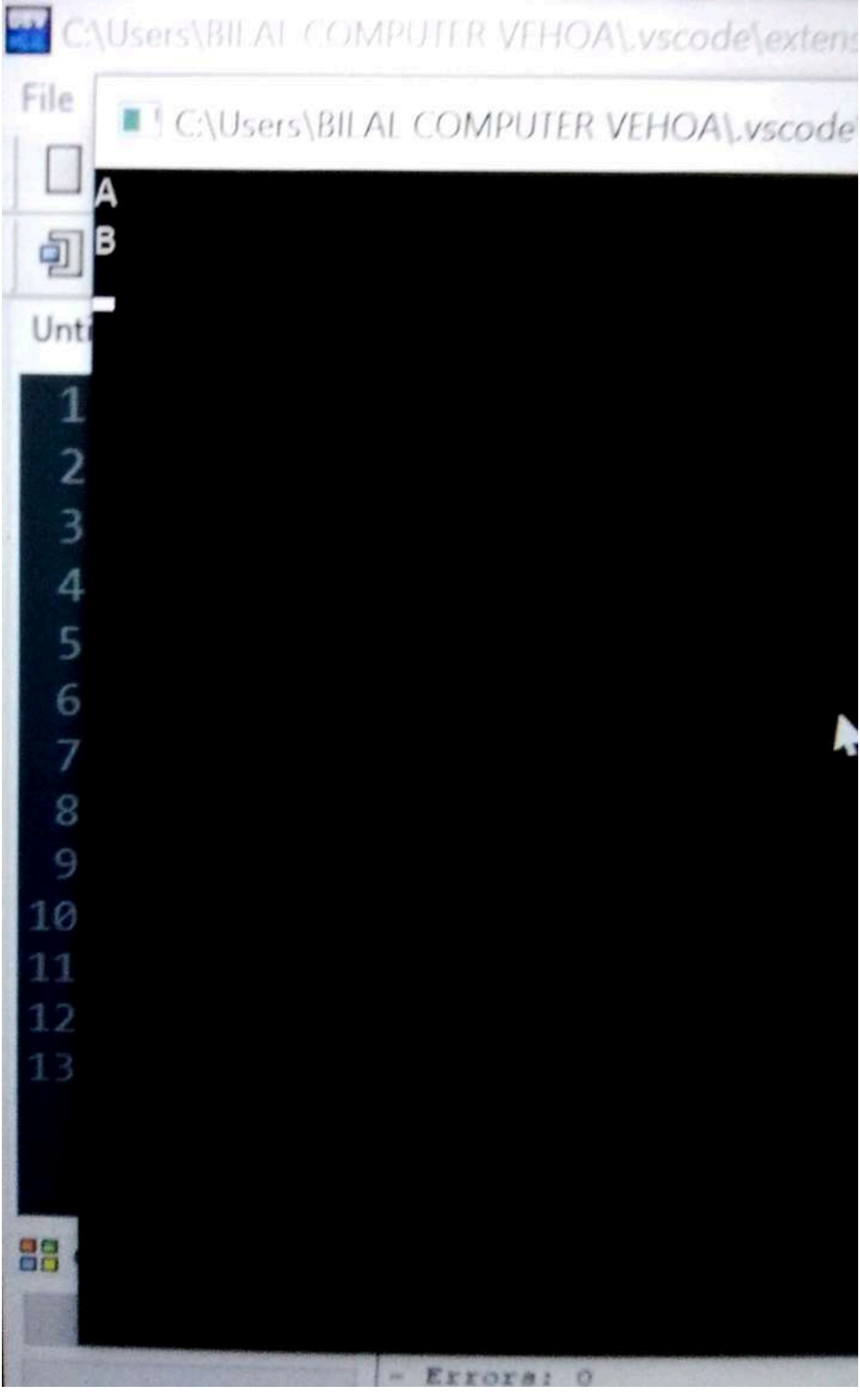
Program 2 :-

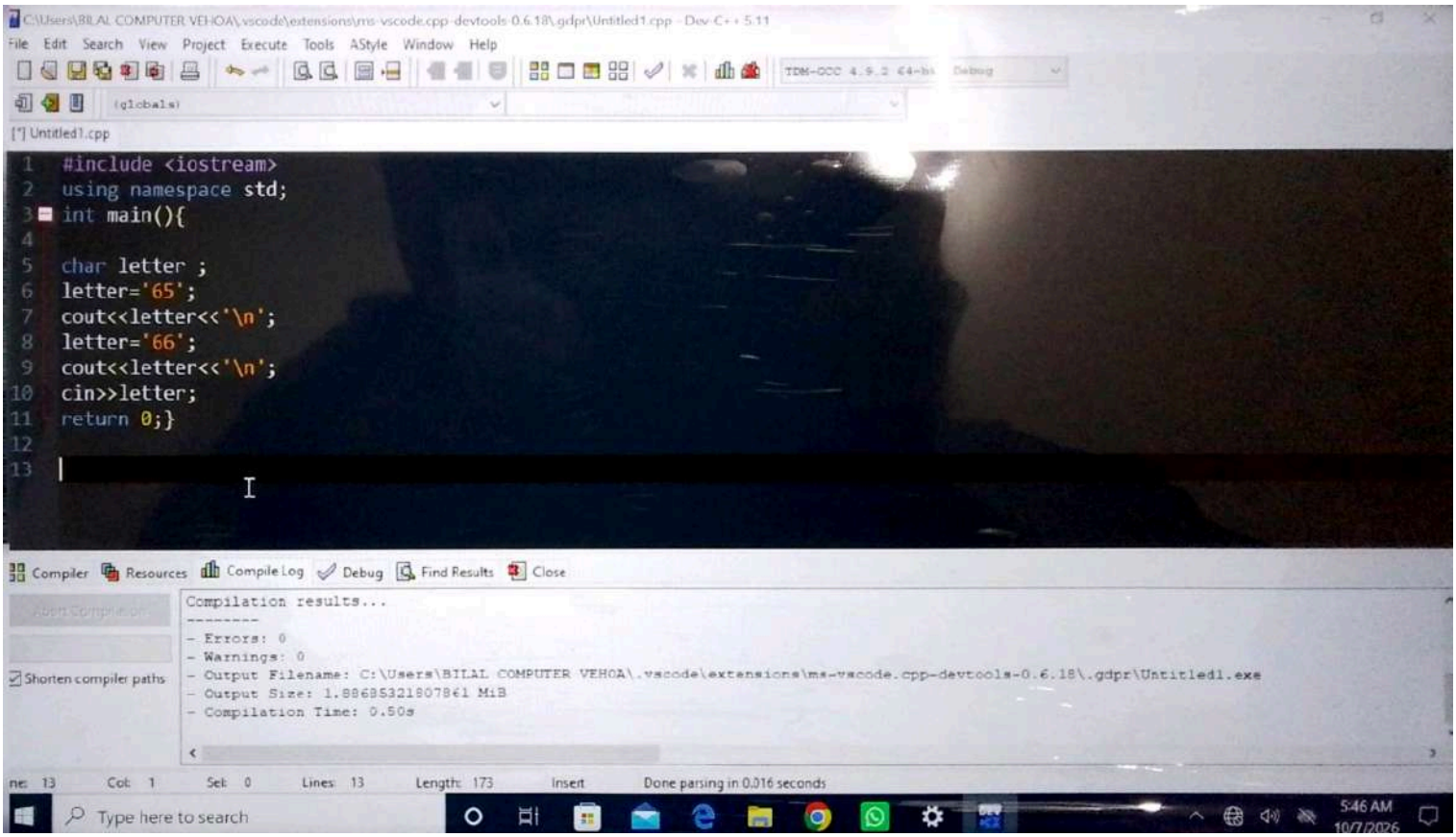
```
1  #include<iostream>
2  using namespace std;
3  int main() {
4  Char Letter;
5  Letter=65;
6  cout<<Letter<<'\n';
7  Letter = 66;
8  cout<<letter<<'\n';
9  cin >> Letter;
10 return 0;}
```

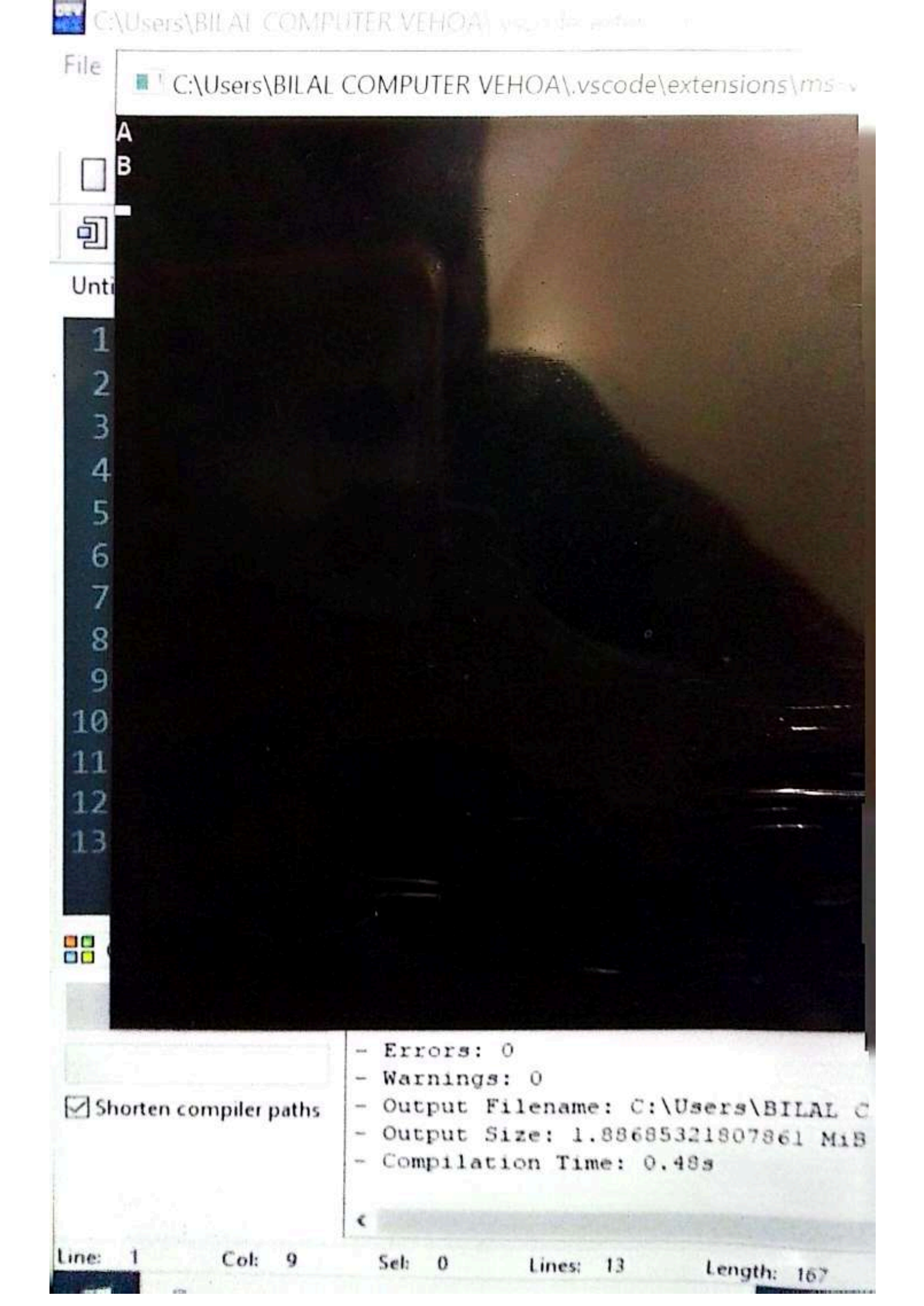
Output :-

```
A
B
```


C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Untitled1.cpp
#include <iostream>
using namespace std;
int main(){
char letter ;
letter='A';
cout<<letter<<'\n';
letter='B';
cout<<letter<<'\n';
cin>>letter;
return 0;
}


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File
C:\Users\BILAL COMPUTER VEHOA\.vscode
A
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11
12
13
- Errors: 0


C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.cpp - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
[*] Untitled1.cpp
#include <iostream>
using namespace std;
int main(){
char letter ;
letter='65';
cout<<letter<<'\n';
letter='66';
cout<<letter<<'\n';
cin>>letter;
return 0;}
Compiler Resources Compile Log Debug Find Results Close
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.exe
- Output Size: 1.88695321807861 MiB
- Compilation Time: 0.50s
Shorten compiler paths
Done parsing in 0.016 seconds
Type here to search
5:46 AM


C:\Users\BILAL COMPUTER VEHOA\
File
C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms
A
B
Unti
1
2
3
4
5
6
7
8
9
10
11
12
13
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL C
- Output Size: 1.88685321807861 MiB
- Compilation Time: 0.48s
Shorten compiler paths
Line: 1
Col: 9
Sel: 0
Lines: 13
Length: 167

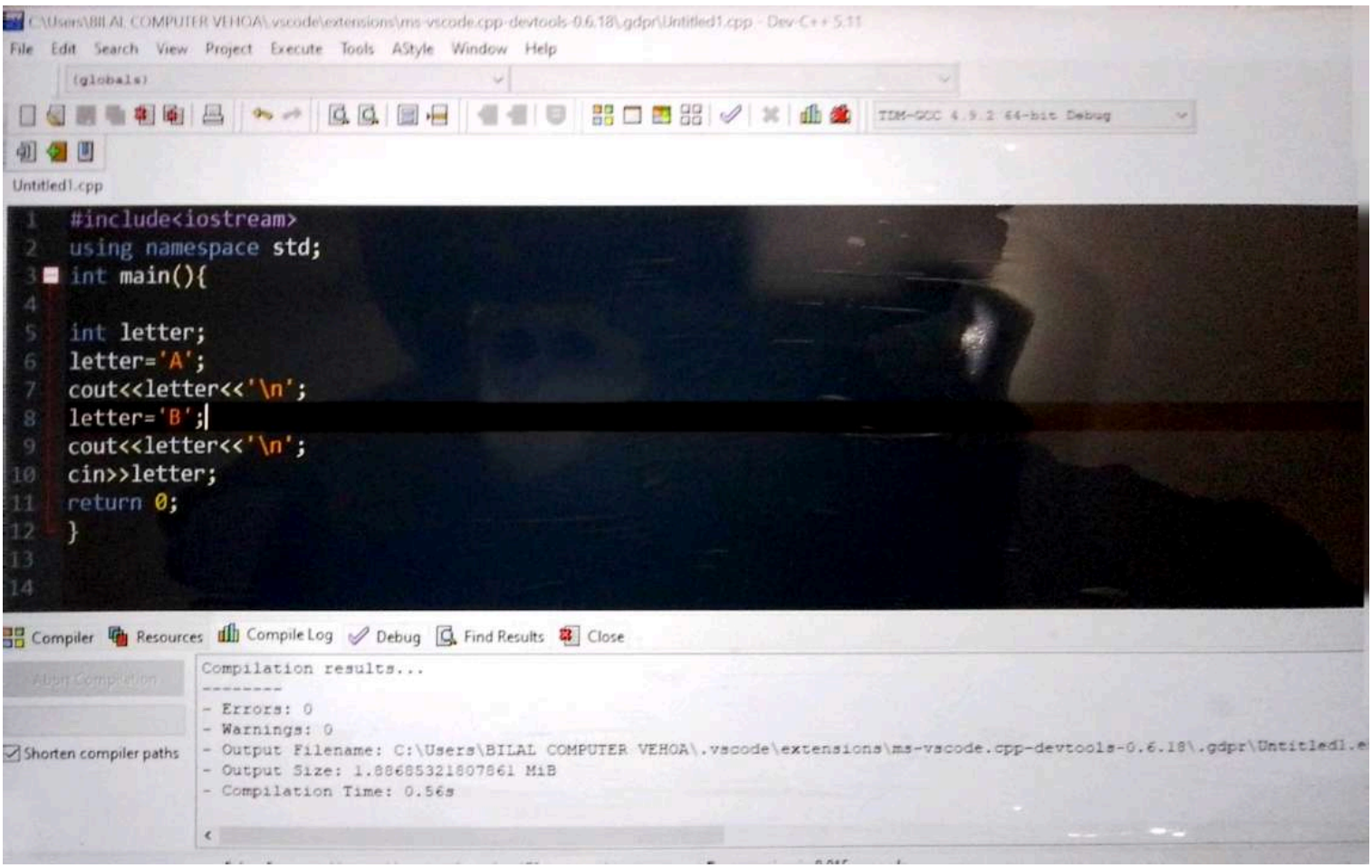
Program 3 :-

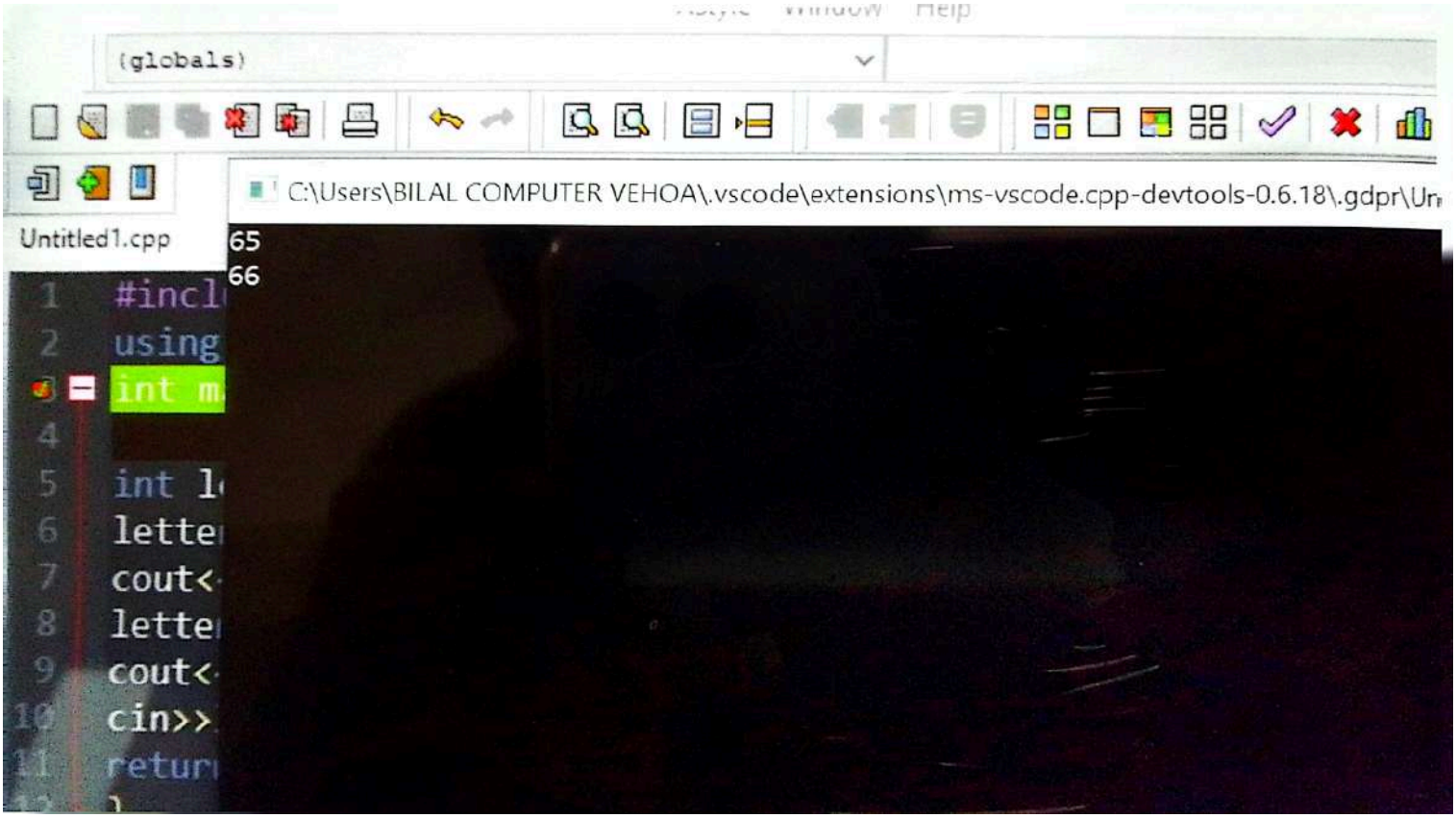
```
1   #include <iostream>
2    Using namespace std;
3    int main() {
4      int Letter;
5      Letter = 'A';
6   Cout << Letter << '\n';
7   Letter = 'B';
8   Cout << Letter << '\n';
9   cin >> letter;
10     return 0; }
```

Output :-

```
65
66
```




C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.cpp - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
TDM-GCC 4.9.2 64-bit Debug
Untitled1.cpp
#include<iostream>
using namespace std;
int main(){
int letter;
letter='A';
cout<<letter<<'\n';
letter='B';
cout<<letter<<'\n';
cin>>letter;
return 0;
}
Compiler Resources Compile Log Debug Find Results Close
Shorten compiler paths
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.e
- Output Size: 1.88685321807861 MiB
- Compilation Time: 0.56s


(globals)
C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Un
Untitled1.cpp
65
66
#incl
using
int m
int l
lette
cout<
lette
cout<
cin>>
retur

Task 2:-

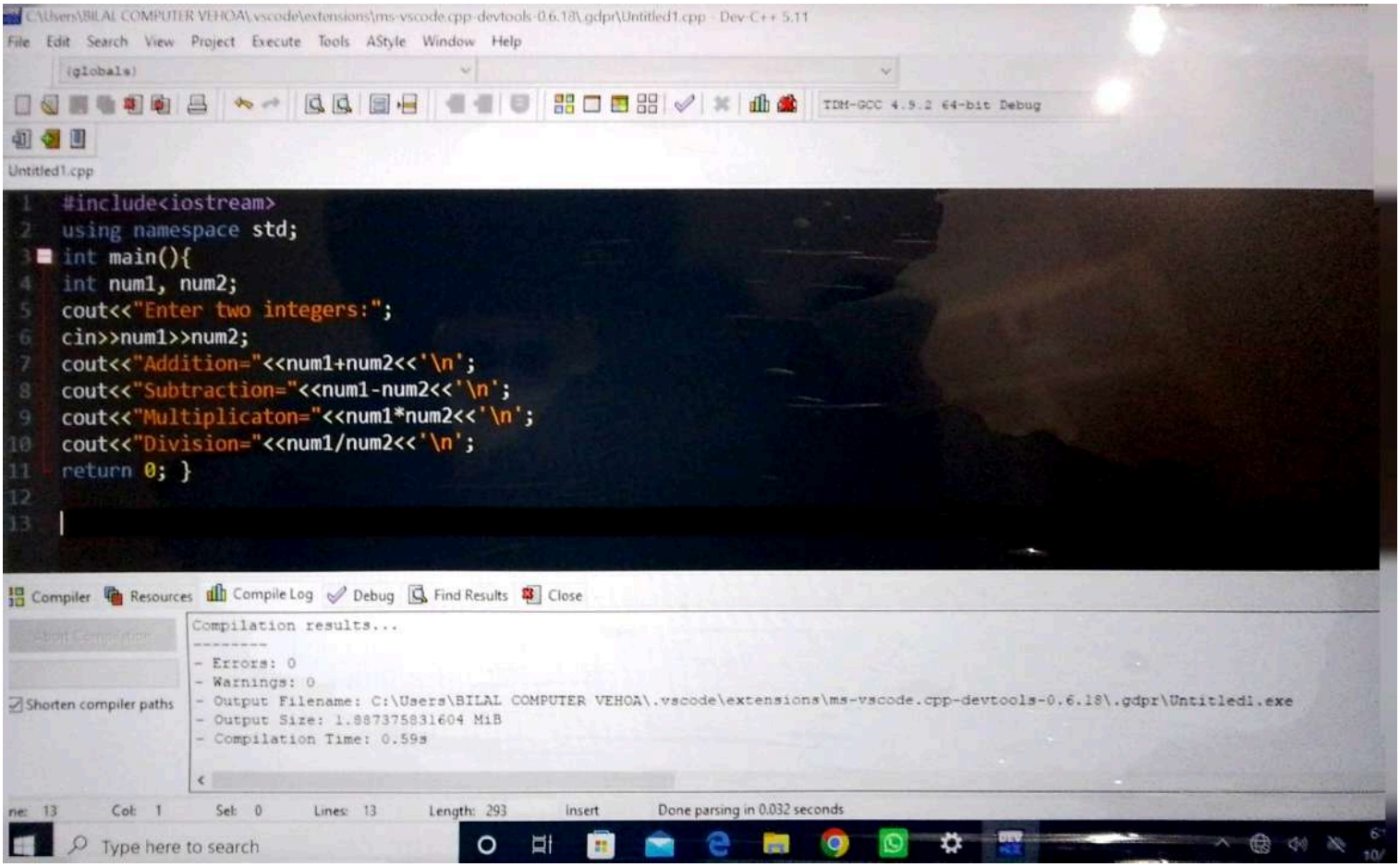
Take two integers as input from the user apply athematic operations on them(+,-,*,/) as print them on Screen.

Program:-

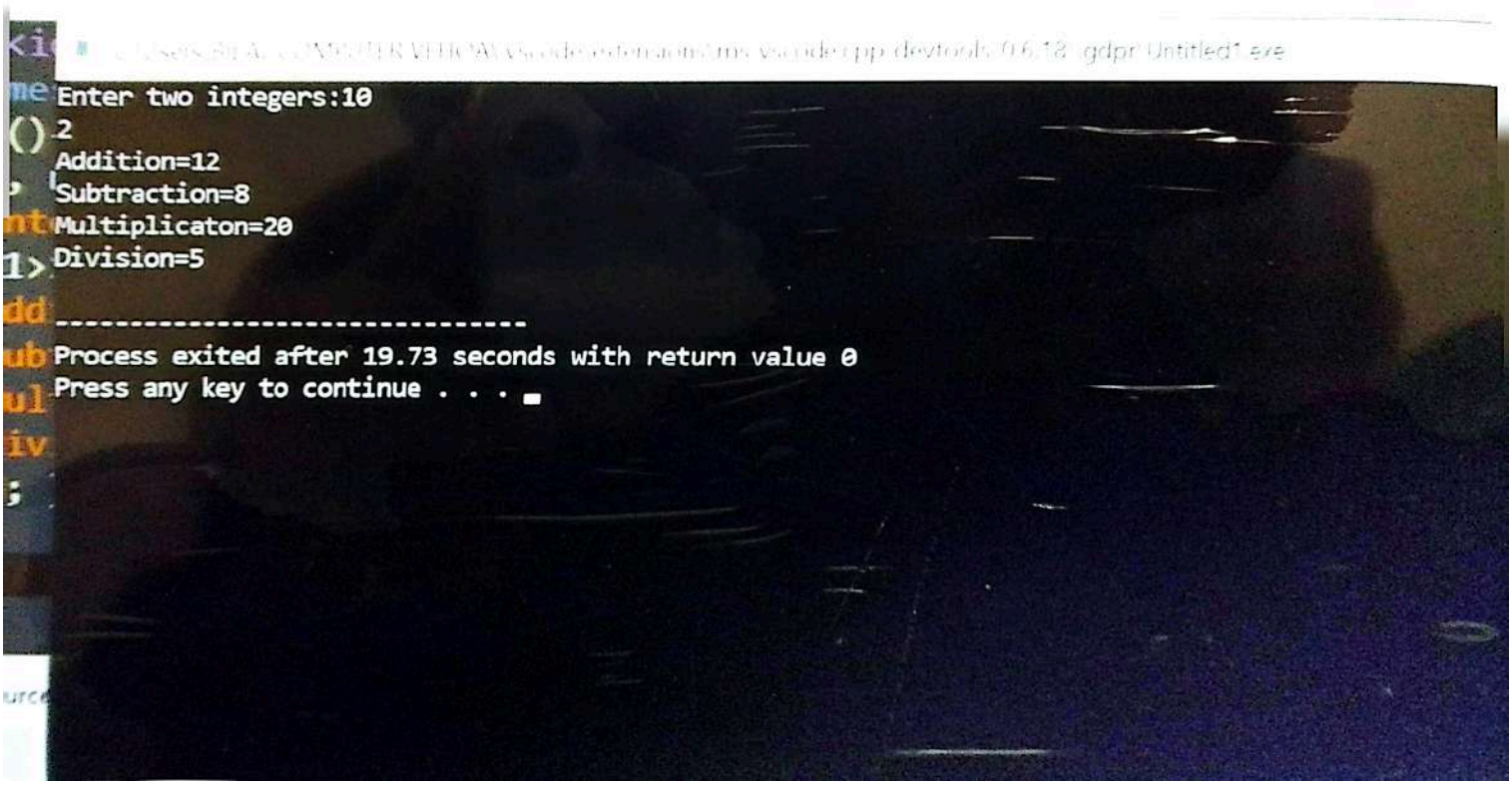
```
1  #include<iostream>
2  using namespace std;
3  int main() {
4     int num1,num2;
5  cout<<"Enter two integers:";
6     cin>>num1>>num2;
7     cout<<"Addition ="<<num1+num2<<'\n';
8  cout<<" Substraction ="<<num1-num2<<'\n';
9  cout<<"Multiplication ="<<num1*num2<<'\n';
10    cout<<"Division= "<<num1/num2<<'\n';
11    return 0;}
```

Output:-

```
Enter two integers :10
 2
Addition =12
Substraction =8
Multiplication=20
Division =5
```


C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.cpp - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
TDM-GCC 4.9.2 64-bit Debug
Untitled1.cpp
#include<iostream>
using namespace std;
int main(){
int num1, num2;
cout<<"Enter two integers:";
cin>>num1>>num2;
cout<<"Addition="<<num1+num2<<'\n';
cout<<"Subtraction="<<num1-num2<<'\n';
cout<<"Multiplicaton="<<num1*num2<<'\n';
cout<<"Division="<<num1/num2<<'\n';
return 0; }
Compiler Resources Compile Log Debug Find Results Close
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.exe
- Output Size: 1.887375831604 MiB
- Compilation Time: 0.59s
Shorten compiler paths
ne: 13 Col: 1 Sel: 0 Lines: 13 Length: 293 Insert Done parsing in 0.032 seconds
Type here to search


Enter two integers:10
2
Addition=12
Subtraction=8
Multiplicaton=20
Division=5

Process exited after 19.73 seconds with return value 0
Press any key to continue . . .

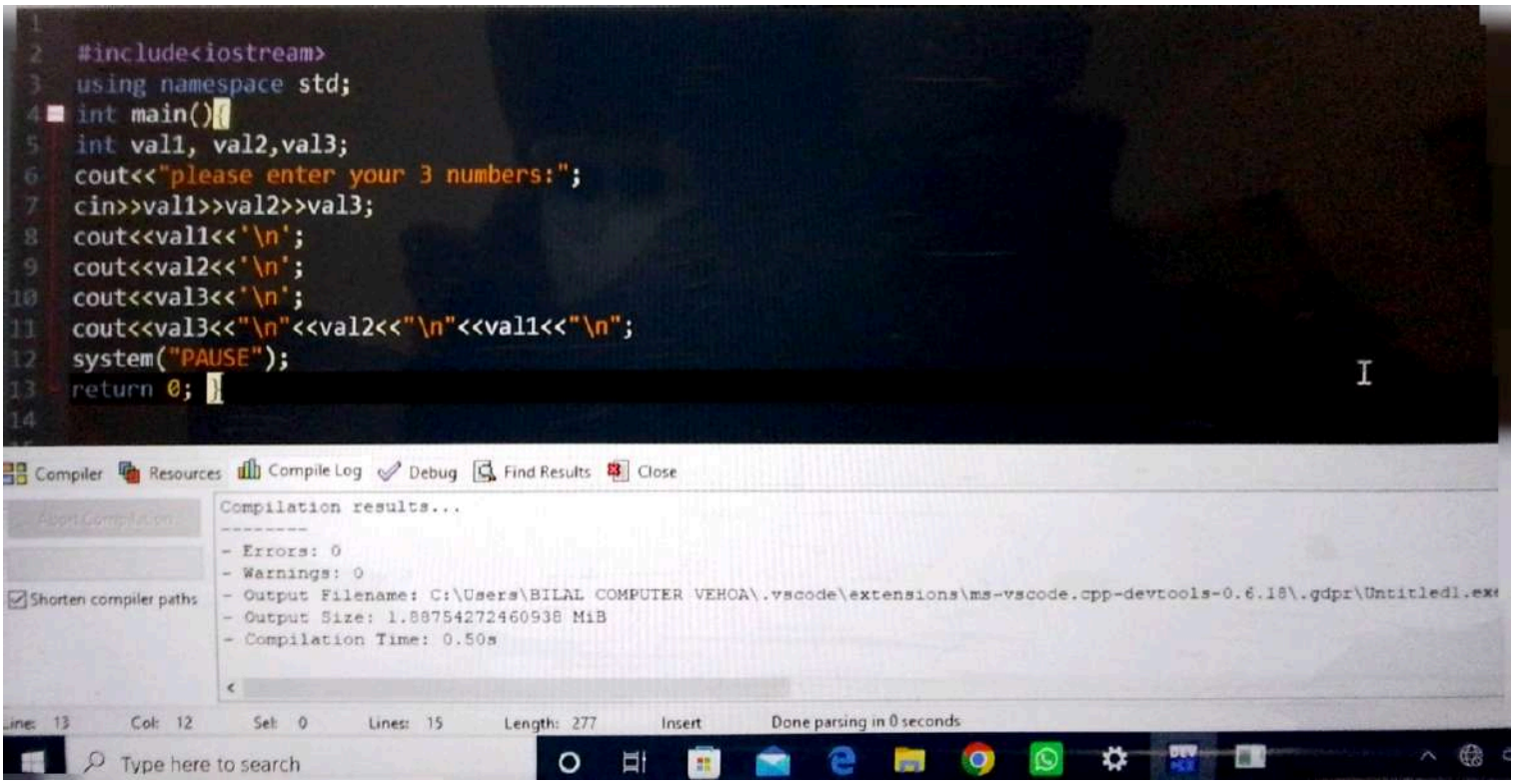
Task 3:-

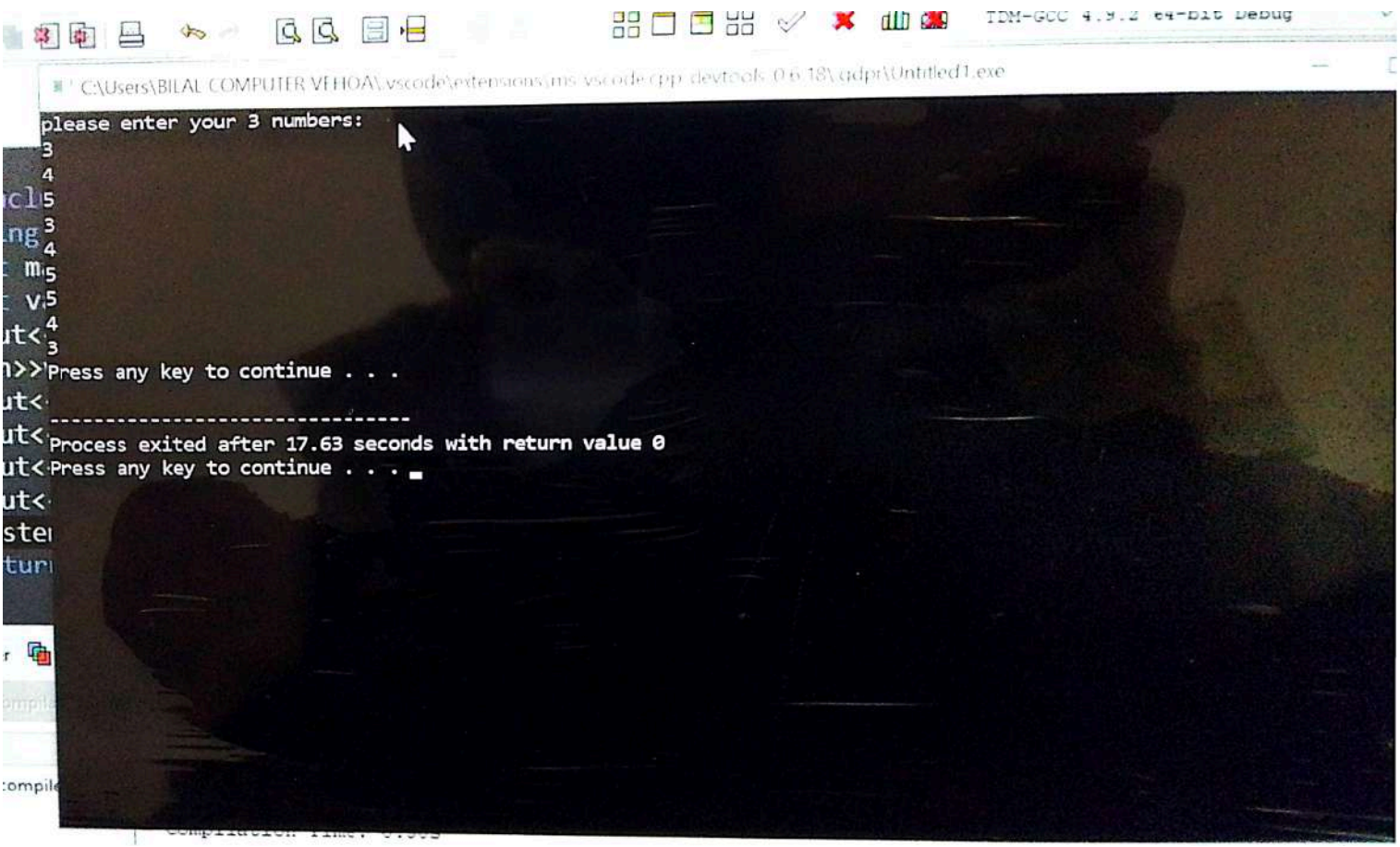
Write the output of the given code

```
#include<iostream>
Using namespace std;
int main(){
int val1, val2, val3;
cout<<"Please enter your 3 numbers: ";
cin>>val1>>val2>>val3;
cout<<val1<<'\n';
cout<<val2<<'\n';
cout<<val3<<'\n';
cout<<val3<<"\n"<<val2"\n"<<val1<<"\n";
system("PAUSE");
return 0 }
```

Output :-

```
Please enter your 3 numbers:
3
4
5
3
4
5
5
4
3
```


#include<iostream>
using namespace std;
int main(){
int val1, val2,val3;
cout<<"please enter your 3 numbers:";
cin>>val1>>val2>>val3;
cout<<val1<<'\n';
cout<<val2<<'\n';
cout<<val3<<'\n';
cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
system("PAUSE");
return 0; }
Compiler
Resources
Compile Log
Debug
Find Results
Close
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.ex
- Output Size: 1.88754272460938 MiB
- Compilation Time: 0.50s
Shorten compiler paths
Line: 13
Col: 12
Sel: 0
Lines: 15
Length: 277
Insert
Done parsing in 0 seconds
Type here to search


C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpptools-0.6.18\.gdpr\Untitled1.exe
please enter your 3 numbers:
3
4
5
3
4
5
5
4
3
Press any key to continue . . .

Process exited after 17.63 seconds with return value 0
Press any key to continue . . .

Task 4 :-

Write a C++ program to calculate the distance between the two points. Note : x1, y1, x2, y2 are all double values.

Formula:

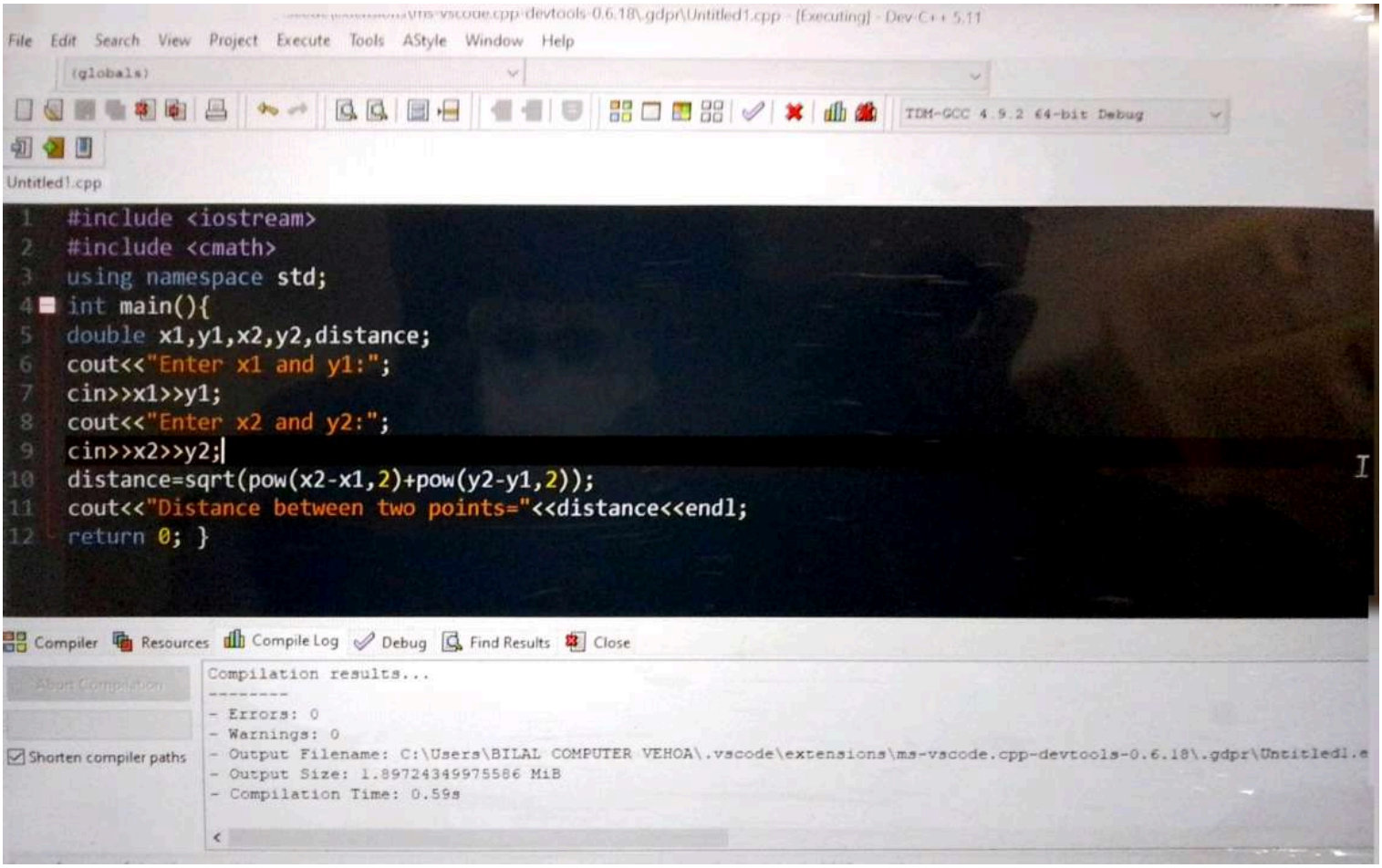
$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

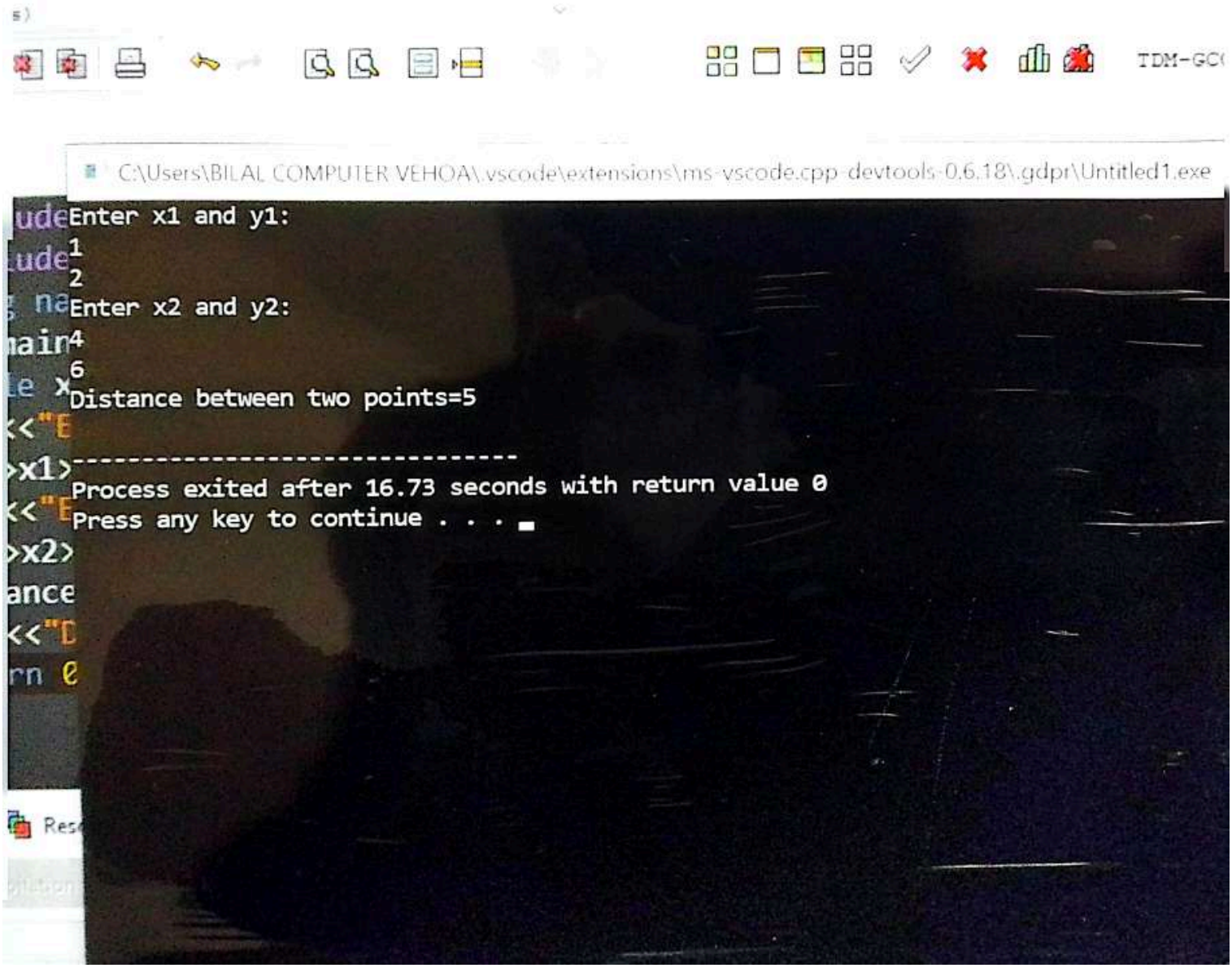
Program :-

```
#Include<iostream>
# include <cmath>
Using namespace std;
int main() {
double x1, y1, x2, y2, distance;
cout << "Enter x1 and y1 :";
cin >>x1>>y1;
cout << "Enter x2 and y2 :";
cin >> x2 >> y2;
distance = sqrt(pow(x2-x1,2)+pow(y2-y1));
cout << "Distance between two points= " << distance<<
endl;
return 0;}
```

Output :-

```
Enter X1 and Y1:
1
2
Enter X2 and Y2:
4
6
Distance between two points =5
```


File Edit Search View Project Execute Tools AStyle Window Help
(globals)
TDM-GCC 4.9.2 64-bit Debug
Untitled1.cpp
#include <iostream>
#include <cmath>
using namespace std;
int main(){
double x1,y1,x2,y2,distance;
cout<<"Enter x1 and y1:";
cin>>x1>>y1;
cout<<"Enter x2 and y2:";
cin>>x2>>y2;
distance=sqrt(pow(x2-x1,2)+pow(y2-y1,2));
cout<<"Distance between two points="<<distance<<endl;
return 0; }
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
Shorten compiler paths
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.e
- Output Size: 1.89724349975586 MiB
- Compilation Time: 0.59s


TDM-GC(
C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.exe
Enter x1 and y1:
1
2
Enter x2 and y2:
4
6
Distance between two points=5

Process exited after 16.73 seconds with return value 0
Press any key to continue . . .

Tasks 5 :-

1. calculate Area of Circle (area = pi*r*r)
2. calculate the Area Rectangle (area = length*width)
3. calculate the Area Triangle (area = base*height*.5)

1. calculate the area of a circle :-

Formula :-

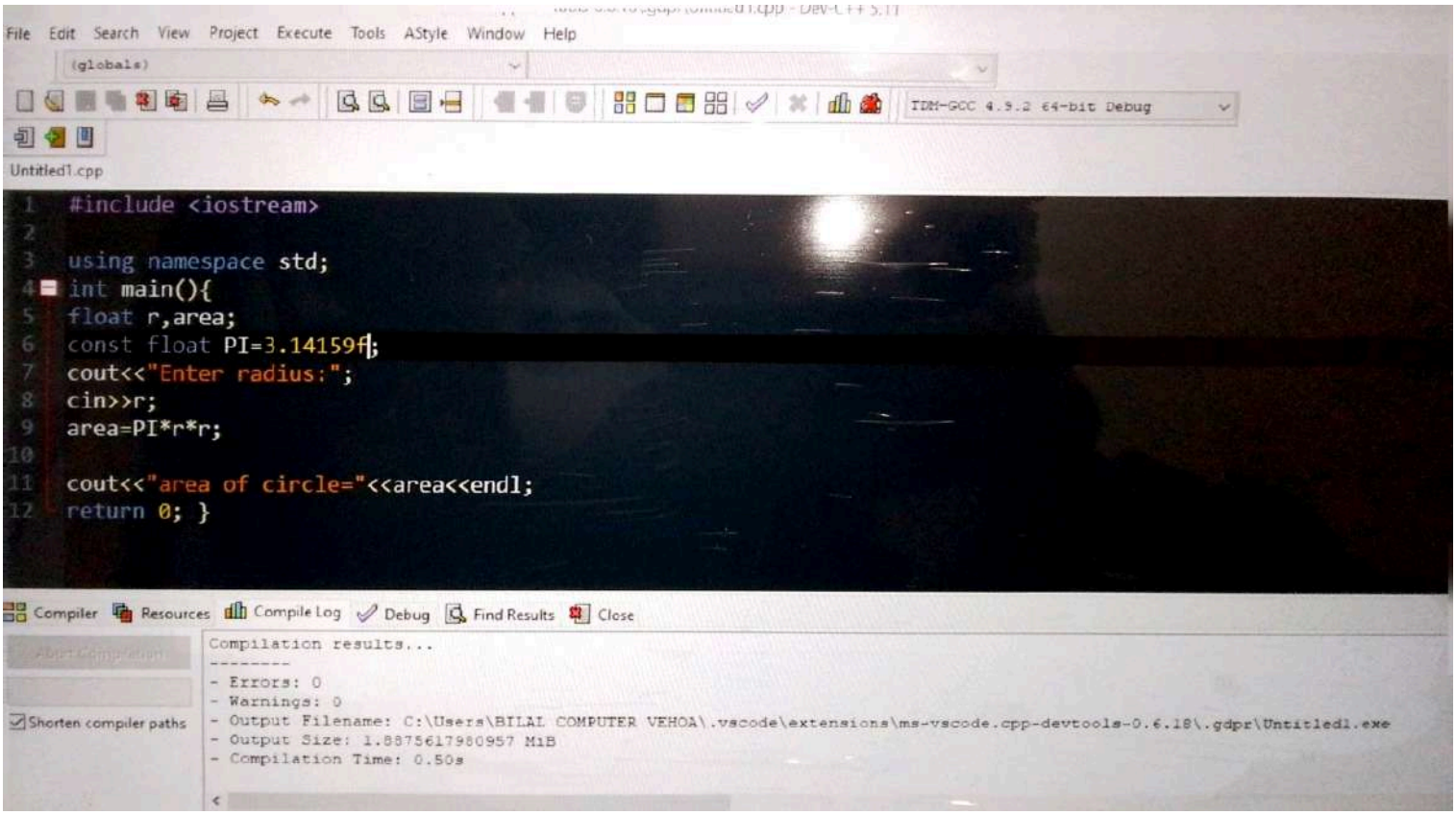
Area = $\pi \times r \times r$

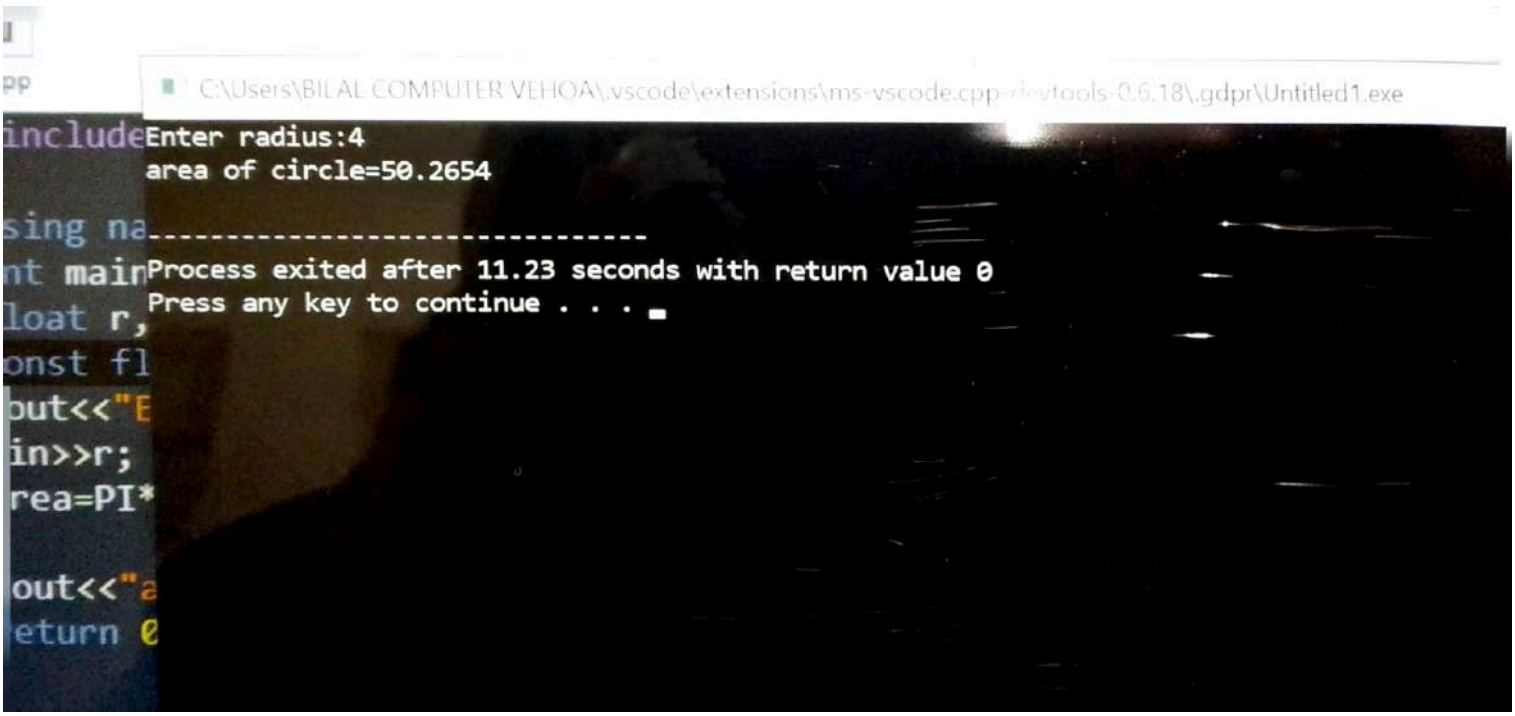
Program:

```
1   #include<iostream>
2   Using namespace std;
3   int main(){
4   float r, area;
5   Const float pi= 3.14159f;
6   Cout<<"Enter radius:";
7     Cin>>r;
8     area = pi*r*r;
9  cout<<" area of circle = "<<area<<endl;
10  return 0;}
```

output :-

```
Enter radius : 4
area of circle = 50.2654
```


File Edit Search View Project Execute Tools AStyle Window Help
(globals)
TDM-GCC 4.9.2 64-bit Debug
Untitled1.cpp
#include <iostream>
using namespace std;
int main(){
float r,area;
const float PI=3.14159f;
cout<<"Enter radius:";
cin>>r;
area=PI*r*r;
cout<<"area of circle="<<area<<endl;
return 0; }
Compiler Resources Compile Log Debug Find Results Close
Shorten compiler paths
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.exe
- Output Size: 1.8875617980957 MiB
- Compilation Time: 0.50s


C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Untitled1.exe
Enter radius:4
area of circle=50.2654

Process exited after 11.23 seconds with return value 0
Press any key to continue . . .

2- Calculate the Area of Rectangle :-

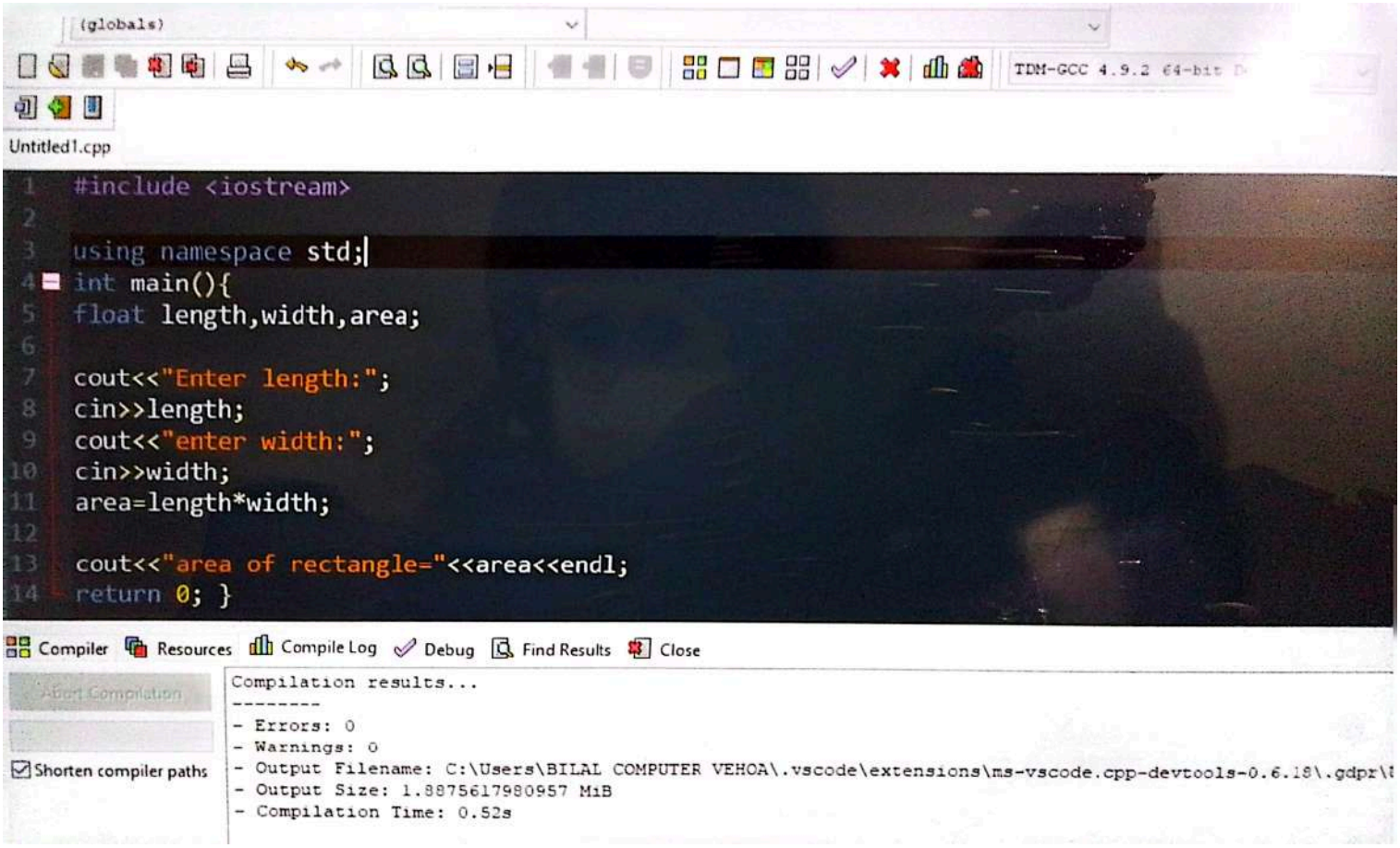
Formula :-

Area = Length X width

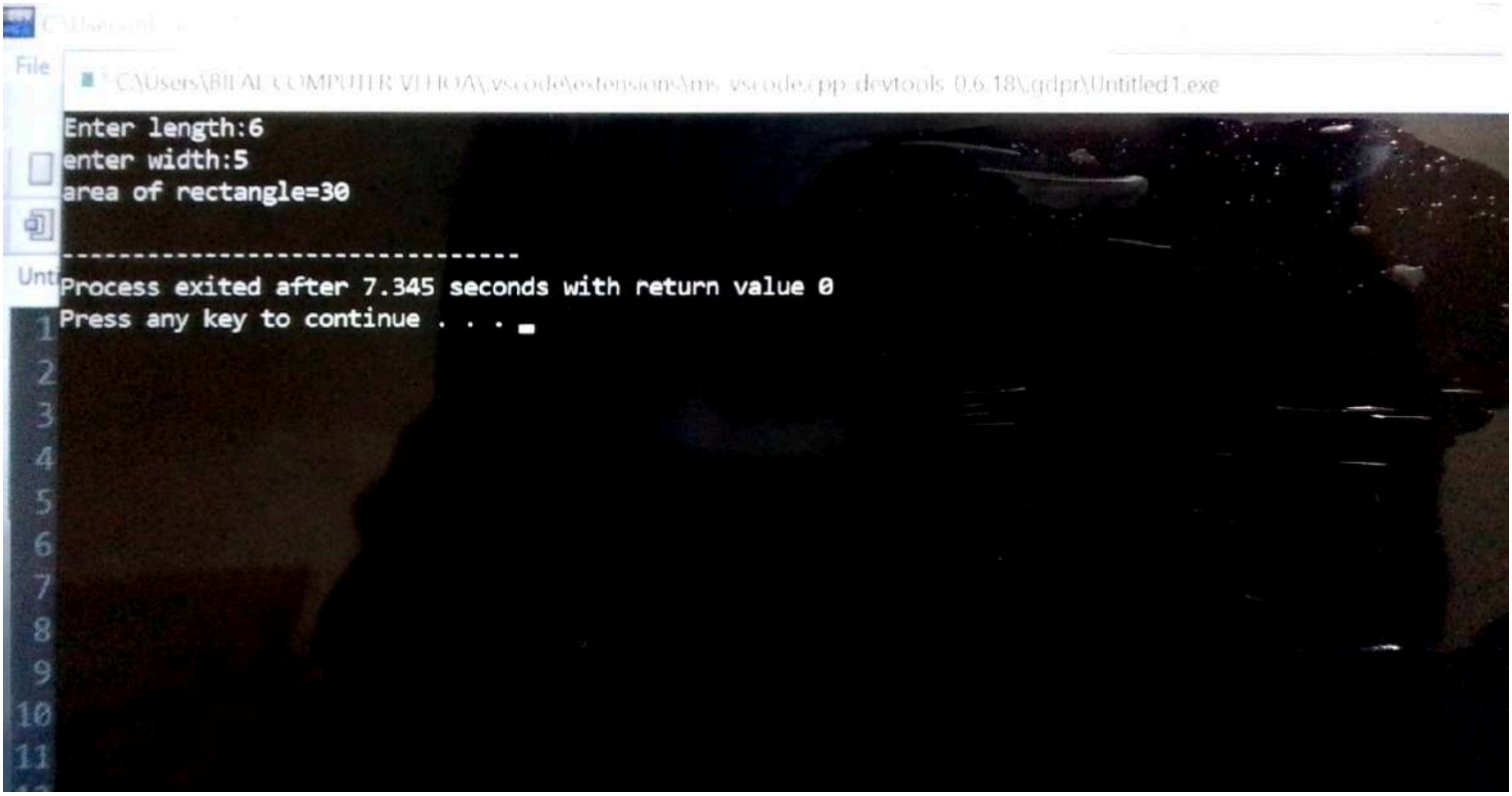
```
1   #include<iostream>
2   Using namespace std;
3    int main(){
4    float length,width,area ;
5   cout<<"Enter length:";
6    cin>>length;
7     cout<<enter width:";
8    Cin>>width;
9    area = length*width;
10   cout<<"area of Rectangle ="<<area<<endl;
11    return 0;}
```

Output :-

Enter length : 6
Enter width : 5
area of Rectangle=30


(globals)
TDM-GCC 4.9.2 64-bit
Untitled1.cpp
#include <iostream>
using namespace std;
int main(){
float length,width,area;
cout<<"Enter length:";
cin>>length;
cout<<"enter width:";
cin>>width;
area=length*width;
cout<<"area of rectangle="<<area<<endl;
return 0; }
Compiler
Resources
Compile Log
Debug
Find Results
Close
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\
- Output Size: 1.8875617980957 MiB
- Compilation Time: 0.52s


File
C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpptools-1.6.18\.gdpr\Untitled1.exe
Enter length:6
enter width:5
area of rectangle=30

Process exited after 7.345 seconds with return value 0
Press any key to continue . . .

3. calculate the area of a Triangle :- (6)

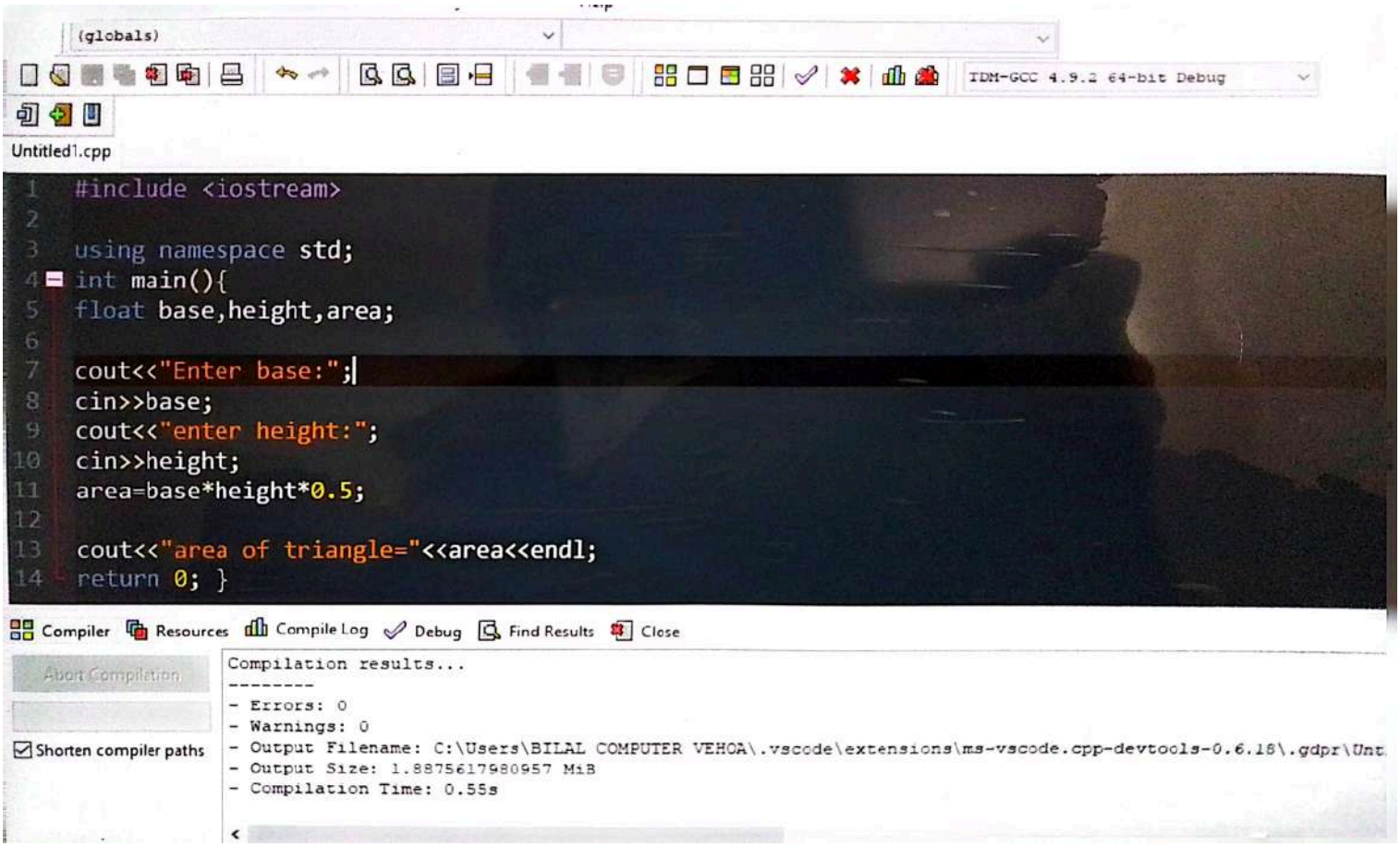
Formula:-

Area = Base X Height X 0.5

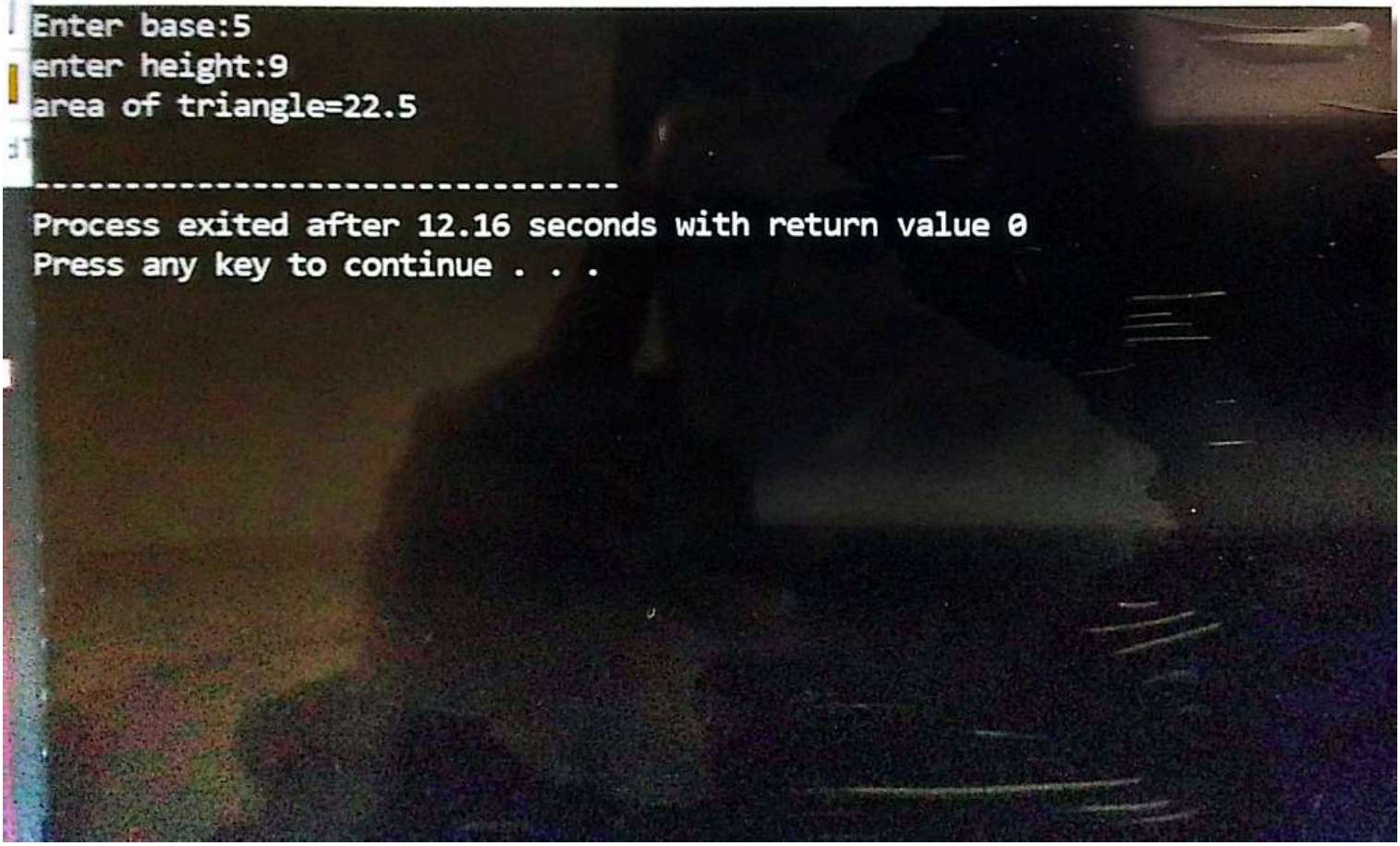
```
1  # include <iostream>
2  Using namespace std;
3  int main() {
4  float base, height, area;
5  cout << "Enter base :";
6  cin >> base;
7  cout << "enter height :";
8  cin >> height;
9  area = base * height * 0.5;
10 cout << "area of triangle = " << area << endl;
11 return 0; }
```

Output :-

```
Enter base : 5
Enter height : 9
area of triangle = 22.5
```


(globals)
IDM-GCC 4.9.2 64-bit Debug
Untitled1.cpp
#include <iostream>
using namespace std;
int main(){
float base,height,area;
cout<<"Enter base:";
cin>>base;
cout<<"enter height:";
cin>>height;
area=base*height*0.5;
cout<<"area of triangle="<<area<<endl;
return 0; }
Compiler
Resources
Compile Log
Debug
Find Results
Close
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.16\.gdpr\Unt
- Output Size: 1.8875617980957 MiB
- Compilation Time: 0.55s


C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gdpr\Un
Enter base:5
enter height:9
area of triangle=22.5

Process exited after 12.16 seconds with return value 0
Press any key to continue . . .

Task 6:- Total purchase

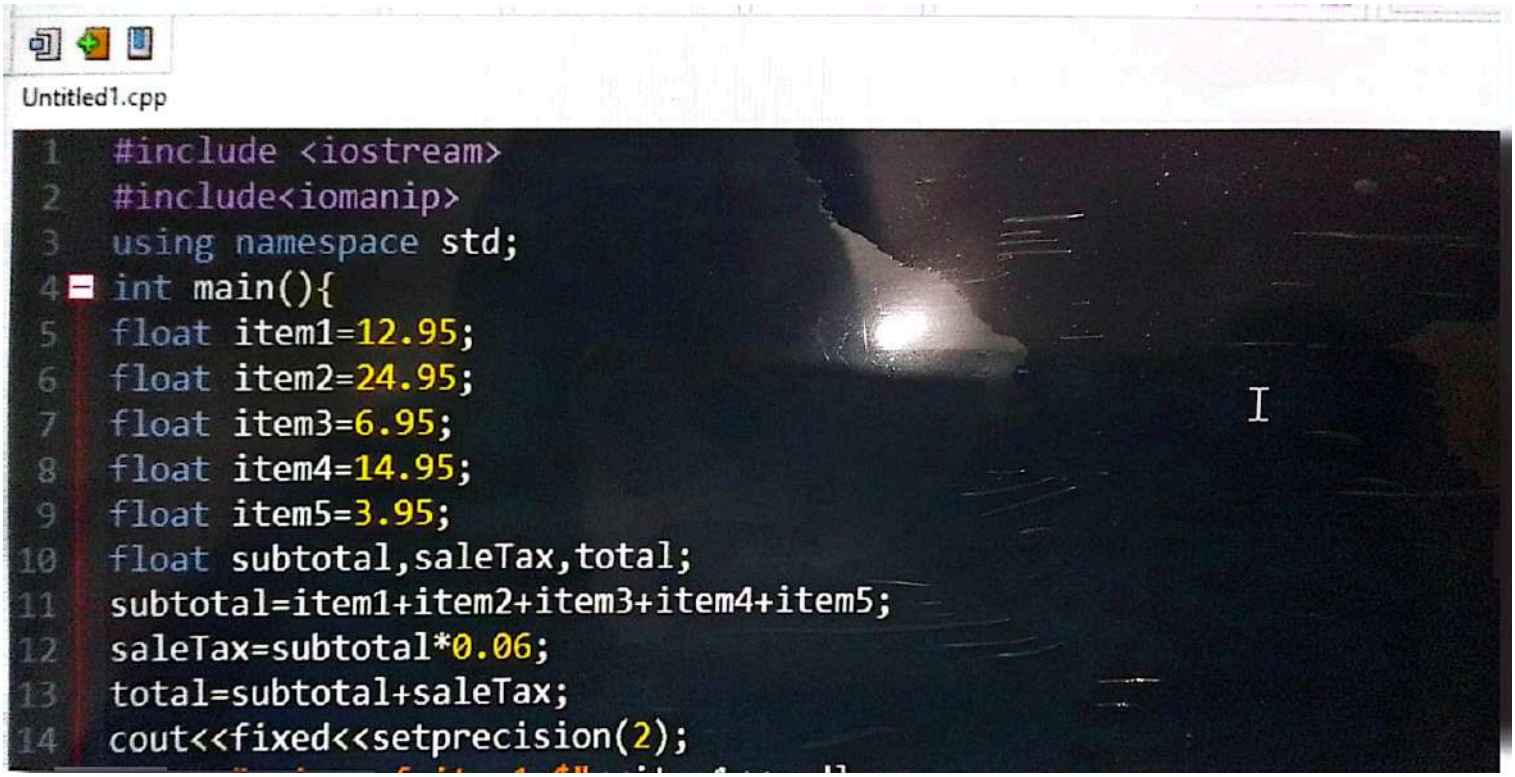
Given prices:-

item 1 = $ 12.95
item 2 = $ 24.95
item 3 = $ 6.95
item 4 = $ 14.95
item 5 = $ 3.95
Sales Tax = 6%.

output:-

price of item 1 = $12.95
price of item 2 = $24.95
price of item 3 = $6.95
price of item 4 = $14.95
price of item 5 = $3.95
Subtotal = $63.75
SaleTax = $3.83
Total = $67.58

```
1  #include <iostream>
2  #include <iomanip>
3  using namespace std;
4   int main () {
5  float item 1 =12.95;
6  float item2 = 24.95;
7  float item3 = 6.95;
8  float item4 = 14.95
9  float item5 = 3.95
10 float subtotal, saleTax, total;
11  subtotal = item1+item2+item3+item4+item5;
12  saleTax = subtotal *0.06;
13  Total = subtotal + saleTax;
14  cout << fixed << setprecision(2);
15  cout << "price of item 1 = $" << item1 << endl;
16  cout << "price of item2 = $" << item2 << endl;
17  cout << "price of item3 = $ " << item3 << endl;
18  cout << "price of item 4 = $" << item4 << endl;
19  cout << "price of item 5 = $" << item5 << endl;
20  cout << "subtotal = $" << Subtotal << endl;
21  cout << "saleTax = $" << saleTax << endl;
22  cout << "Total = $ " << total << endl;
23   return 0; }
```

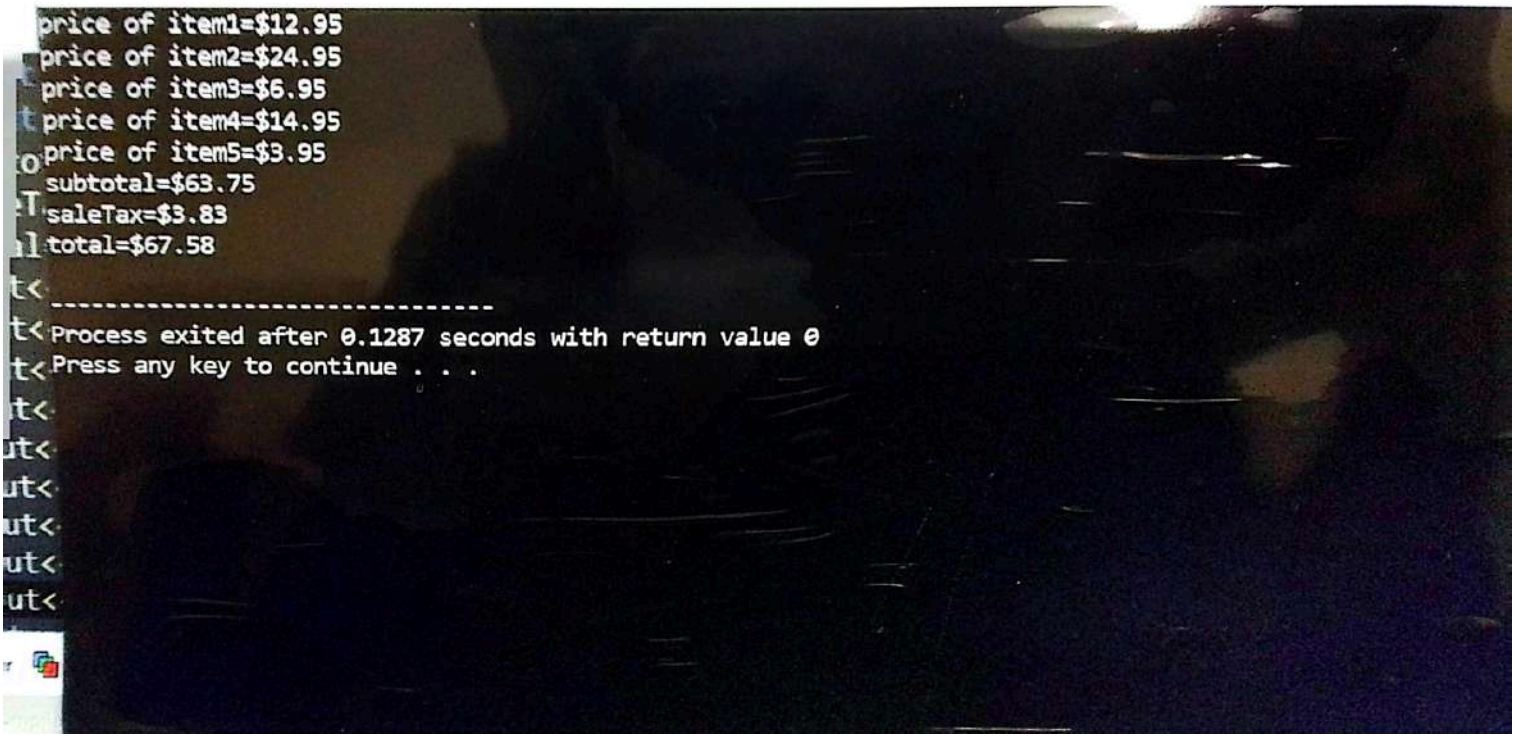

Untitled1.cpp
#include <iostream>
#include<iomanip>
using namespace std;
int main(){
float item1=12.95;
float item2=24.95;
float item3=6.95;
float item4=14.95;
float item5=3.95;
float subtotal,saleTax,total;
subtotal=item1+item2+item3+item4+item5;
saleTax=subtotal*0.06;
total=subtotal+saleTax;
cout<<fixed<<setprecision(2);

```
14  cout<<fixed<<setprecision(2);
15  cout<<"price of item1=$"<<item1<<endl;
16  cout<<"price of item2=$"<<item2<<endl;
17  cout<<"price of item3=$"<<item3<<endl;
18  cout<<"price of item4=$"<<item4<<endl;
19  cout<<"price of item5=$"<<item5<<endl;
20  cout<<"subtotal=$"<<subtotal<<endl;
21  cout<<"saleTax=$"<<saleTax<<endl;
22  cout<<"total=$"<<total<<endl;
23  return 0; }
```

Compiler | Resources | Compile Log | Debug | Find Results | Close

Abort Compilation

```
Processing C++ source file...
--------
- C++ Compiler: %BinDir0%\g++.exe
```


price of item1=$12.95
price of item2=$24.95
price of item3=$6.95
price of item4=$14.95
price of item5=$3.95
subtotal=$63.75
saleTax=$3.83
total=$67.58

Process exited after 0.1287 seconds with return value 0
Press any key to continue . . .

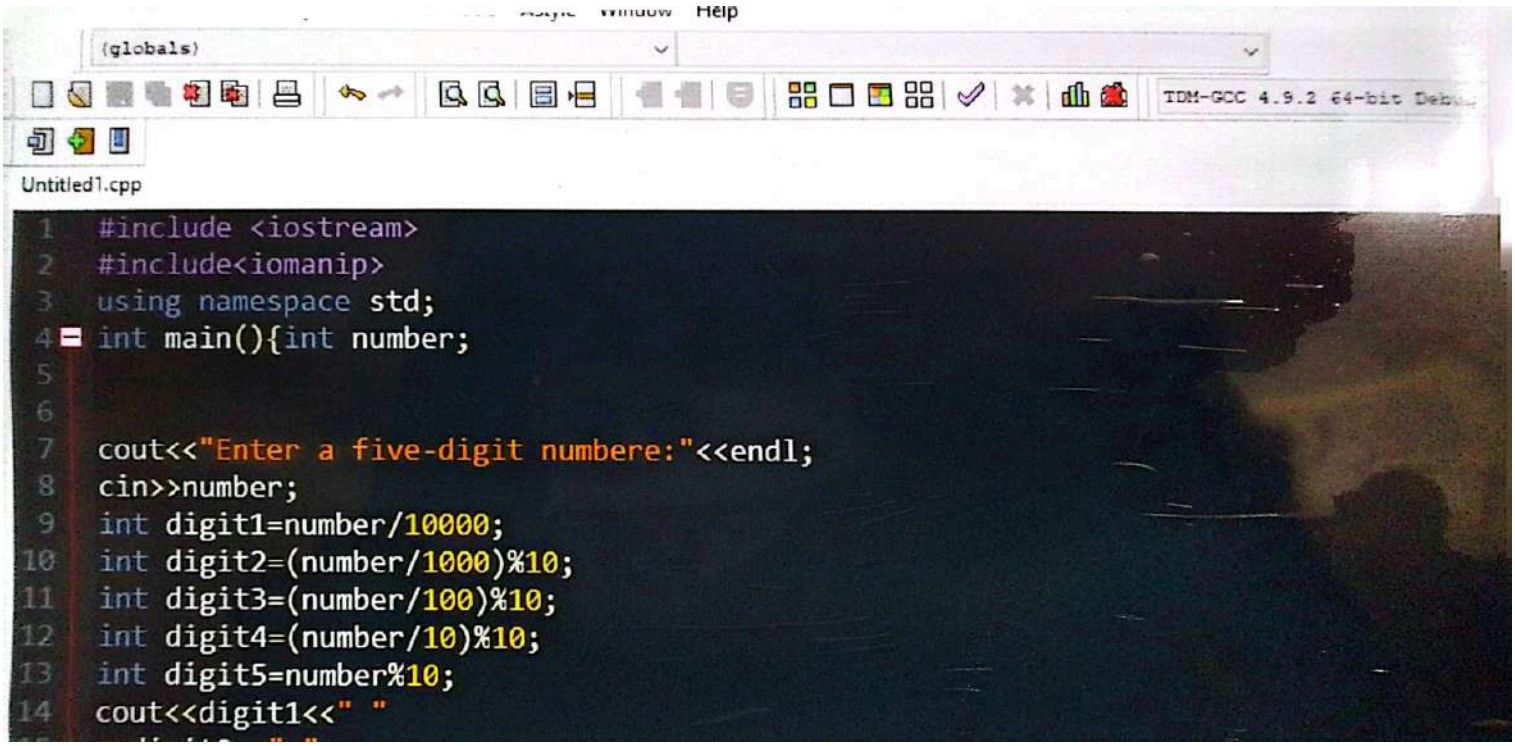
Bonus Task :- Break Number :-

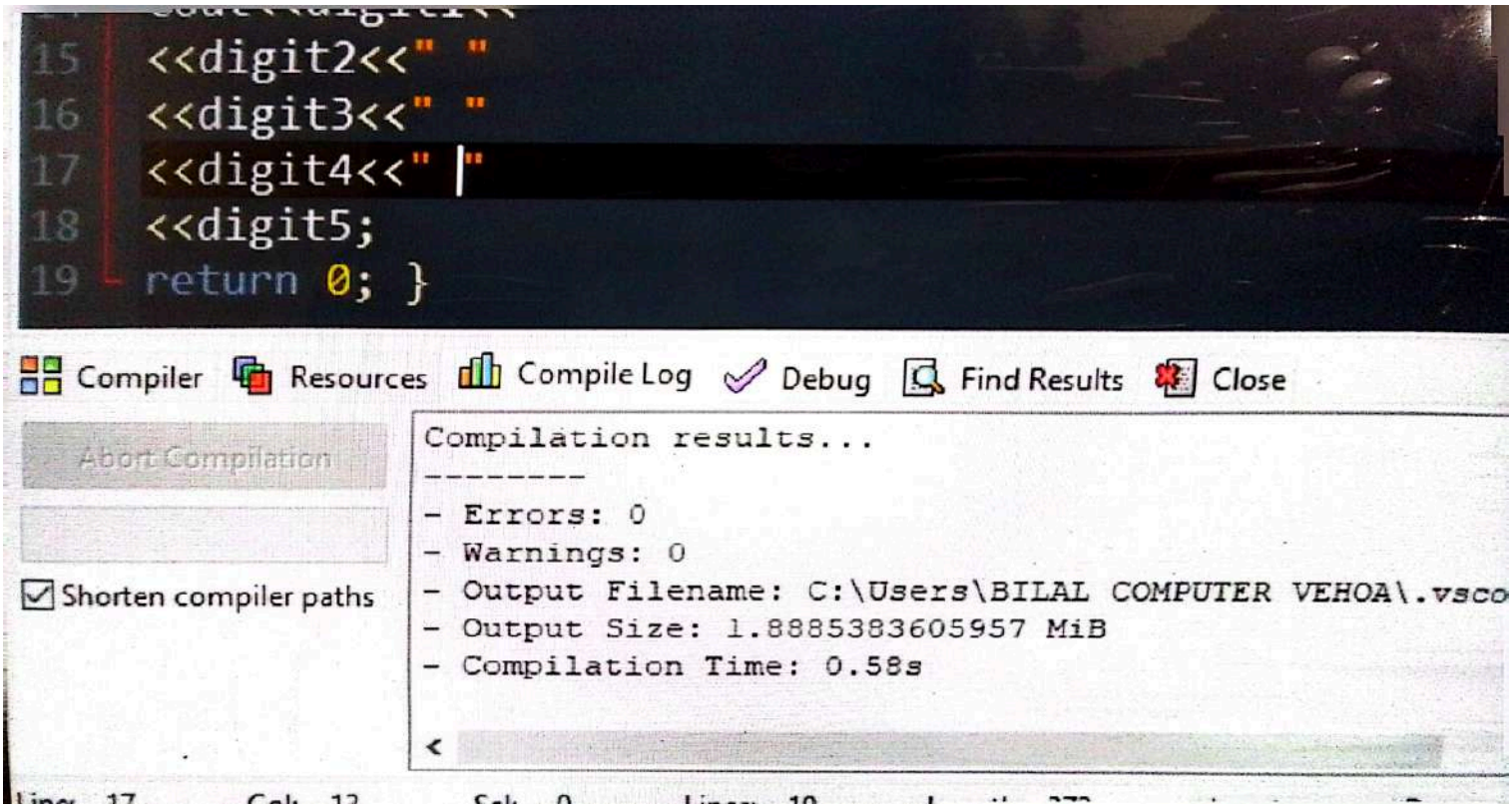
write a program that input a five digit integer and separates it into its indivisiual digits, displaying the digits Separated by three spaces.

```
# include <iostream>
# include <iomanip>
Using namespace std;
int main() { int number;
cout << "Enter a five-digit number:" << endl;
Cin >> number;
int digit1 = number / 10000;
int digit2 = (number / 1000) % 10;
int digit3 = (number / 100) % 10;
int digit4 = (number / 10) % 10;
int digit 5 = number % 10;
 Cout << digit1 << " "
<< digit2 << " "
<< digit3 << " "
<< digit4 << " "
<< digit5;
return 0; }
```

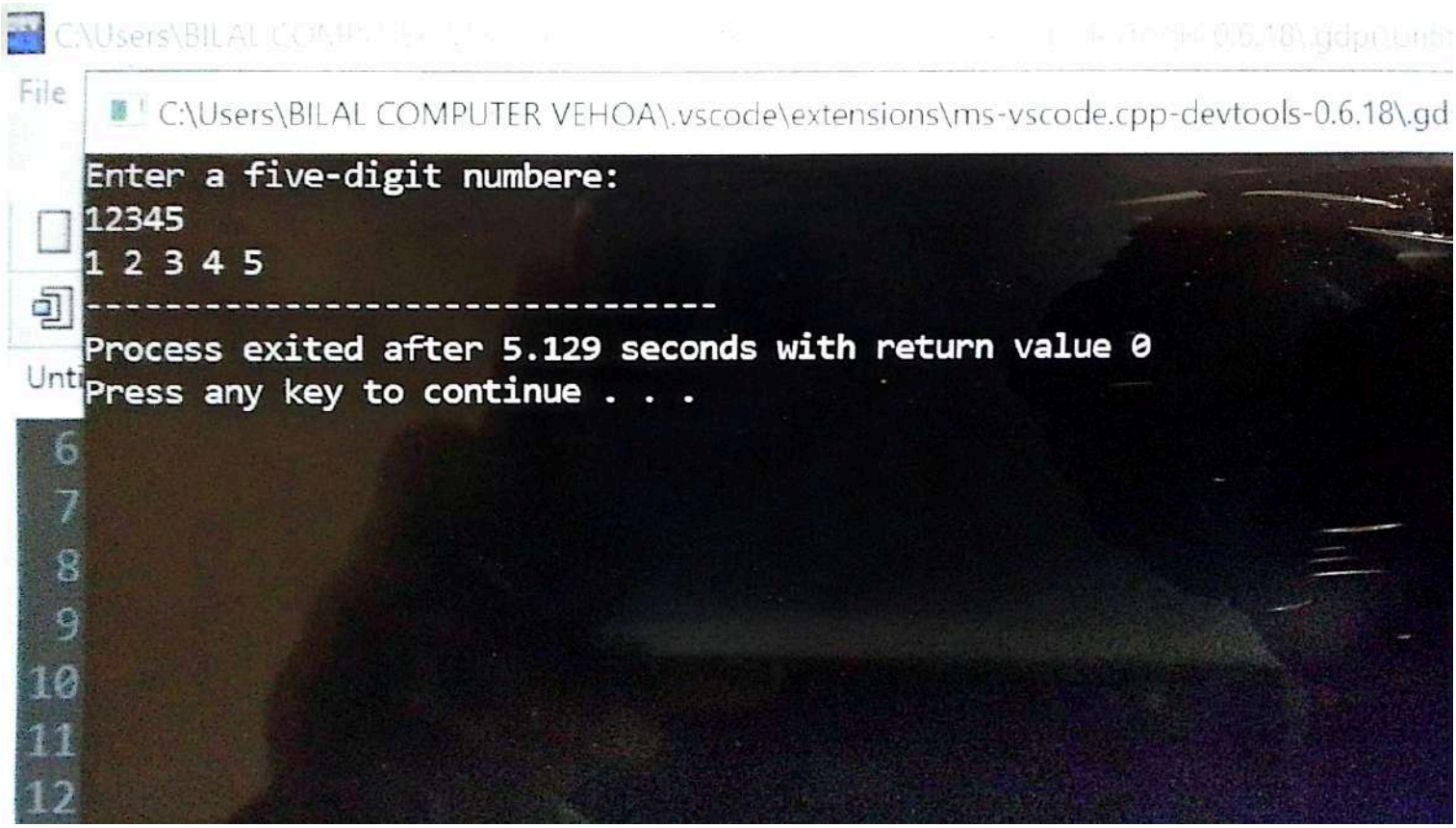
output :-

```
Enter a five digit number:
12345
1 2 3 4 5
```


Window Help
(globals)
TDM-GCC 4.9.2 64-bit Deb
Untitled1.cpp
#include <iostream>
#include<iomanip>
using namespace std;
int main(){int number;
cout<<"Enter a five-digit numbere:"<<endl;
cin>>number;
int digit1=number/10000;
int digit2=(number/1000)%10;
int digit3=(number/100)%10;
int digit4=(number/10)%10;
int digit5=number%10;
cout<<digit1<<" "


15 <<digit2<<" "
16 <<digit3<<" "
17 <<digit4<<" "
18 <<digit5;
19 return 0; }
Compiler
Resources
Compile Log
Debug
Find Results
Close
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vsco
- Output Size: 1.8885383605957 MiB
- Compilation Time: 0.58s


C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-devtools-0.6.18\.gd
Enter a five-digit numbere:
12345
1 2 3 4 5

Process exited after 5.129 seconds with return value 0
Press any key to continue . . .

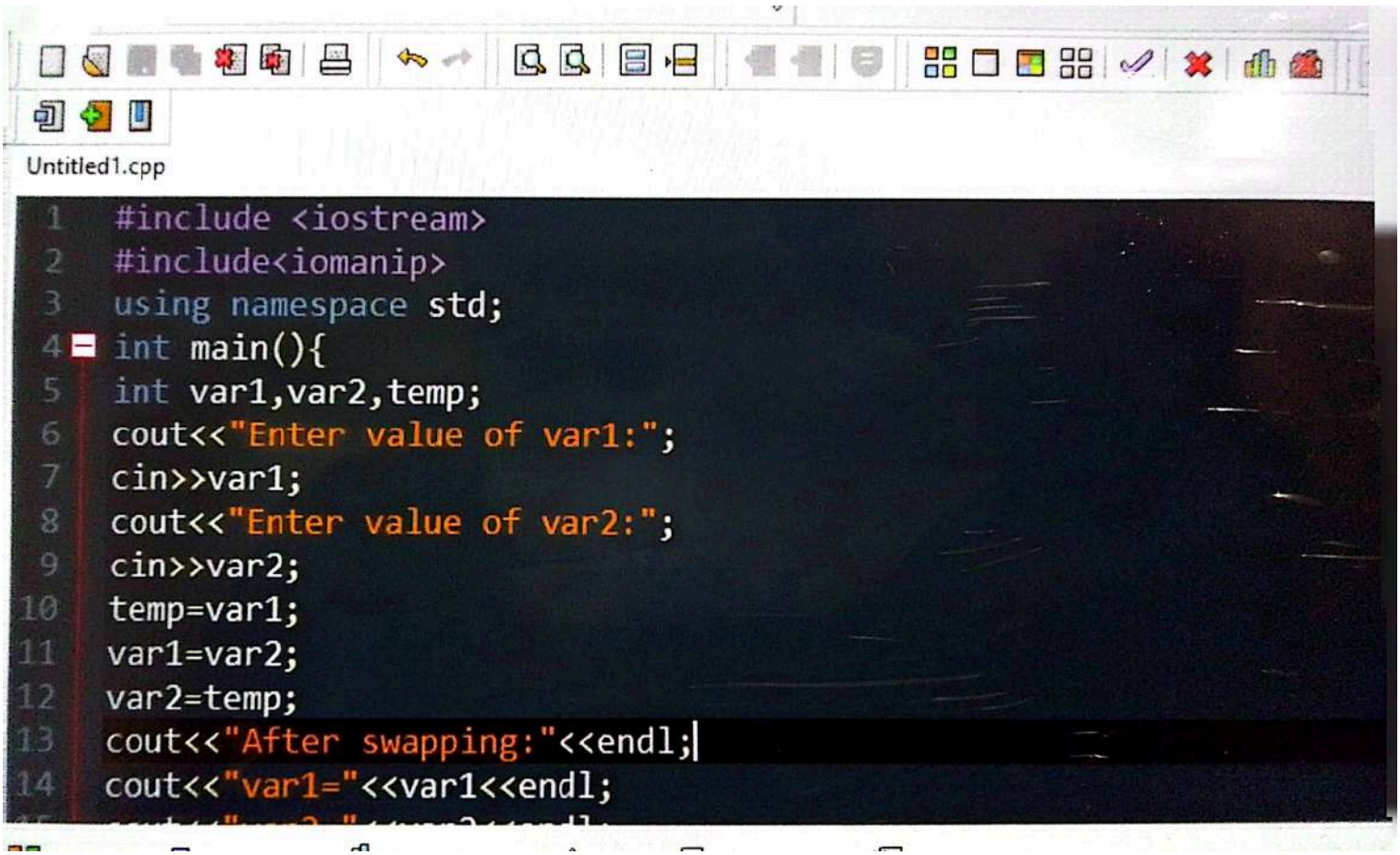
Home Task 1 - Exchange :- (6)

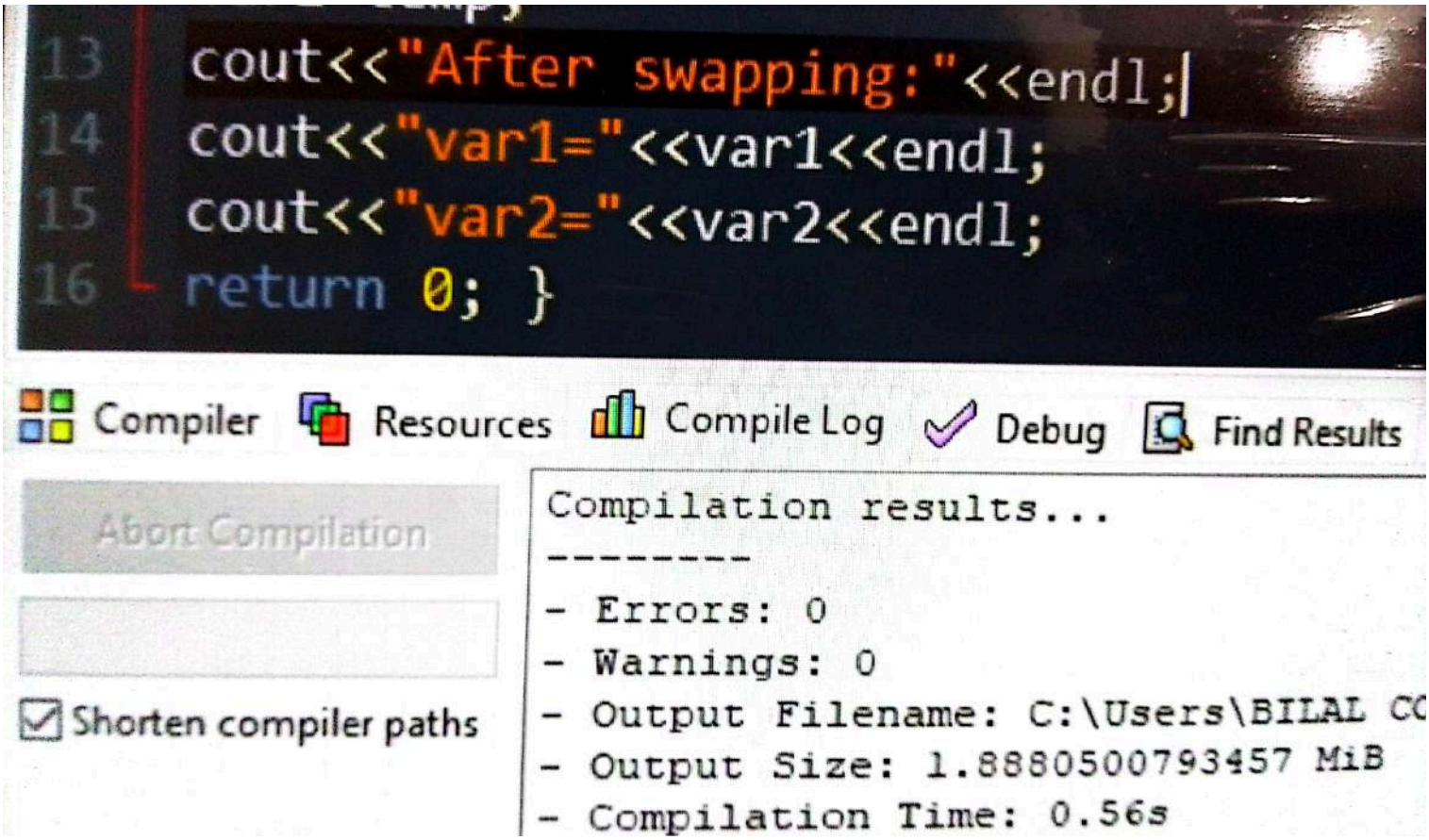
write a paragraph that reads two variables and swaps their values.

```
1  #include<iostream>
2  #include<iomanip>
3  Using namespace std;
4  int main(){
5  int var1, var2, temp;
6  cout<<"Enter value of var1:";
7  cin>>var1;
8  cout<<"Enter value of var2:";
9  cin>>var2;
10  temp=var1;
11  var1 = temp;
12  cout<<"After swapping:"<<endl;
13  cout<<"var1="<<var1<<endl;
14  cout<<"var2="<<var2<<endl;
15  return 0;}
```

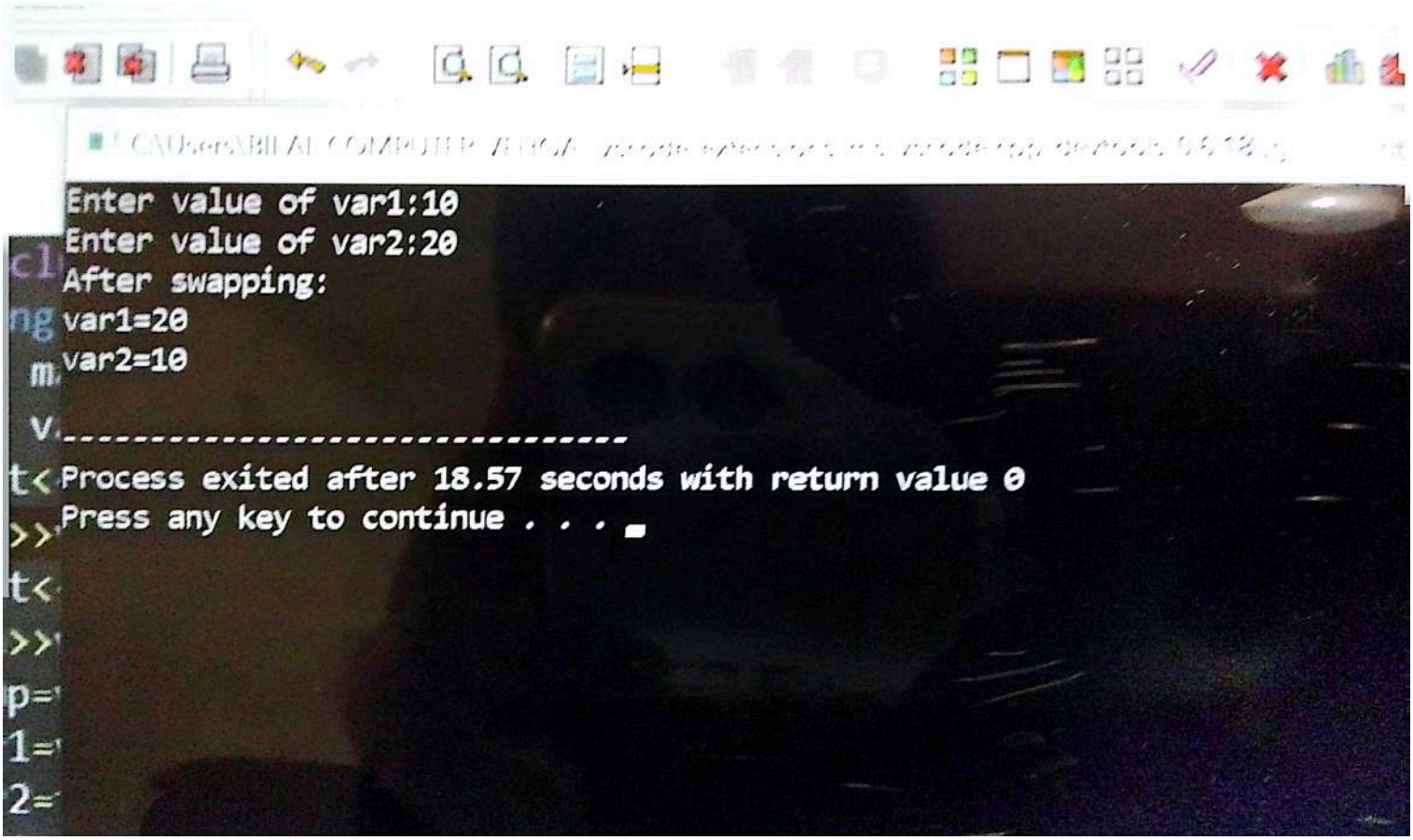
Output :-

Before swapping:
Enter value of var 1 : 10
Enter value of var2 : 20
After swapping:
var1 = 20
var2 = 10


Untitled1.cpp
#include <iostream>
#include<iomanip>
using namespace std;
int main(){
int var1,var2,temp;
cout<<"Enter value of var1:";
cin>>var1;
cout<<"Enter value of var2:";
cin>>var2;
temp=var1;
var1=var2;
var2=temp;
cout<<"After swapping:"<<endl;
cout<<"var1="<<var1<<endl;


13 cout<<"After swapping:"<<endl;
14 cout<<"var1="<<var1<<endl;
15 cout<<"var2="<<var2<<endl;
16 return 0; }
Compiler
Resources
Compile Log
Debug
Find Results
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL CO
- Output Size: 1.8880500793457 MiB
- Compilation Time: 0.56s


Enter value of var1:10
Enter value of var2:20
After swapping:
var1=20
var2=10

Process exited after 18.57 seconds with return value 0
Press any key to continue . . .

Home Task 2- Make Upper:-

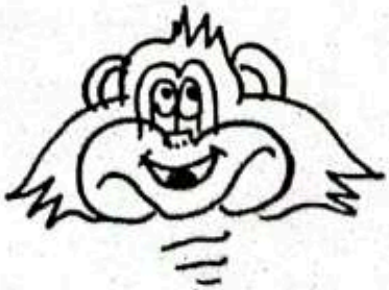
write a paragraph that takes an alphabet in lowercase and displays it in uppercase using ASCII.

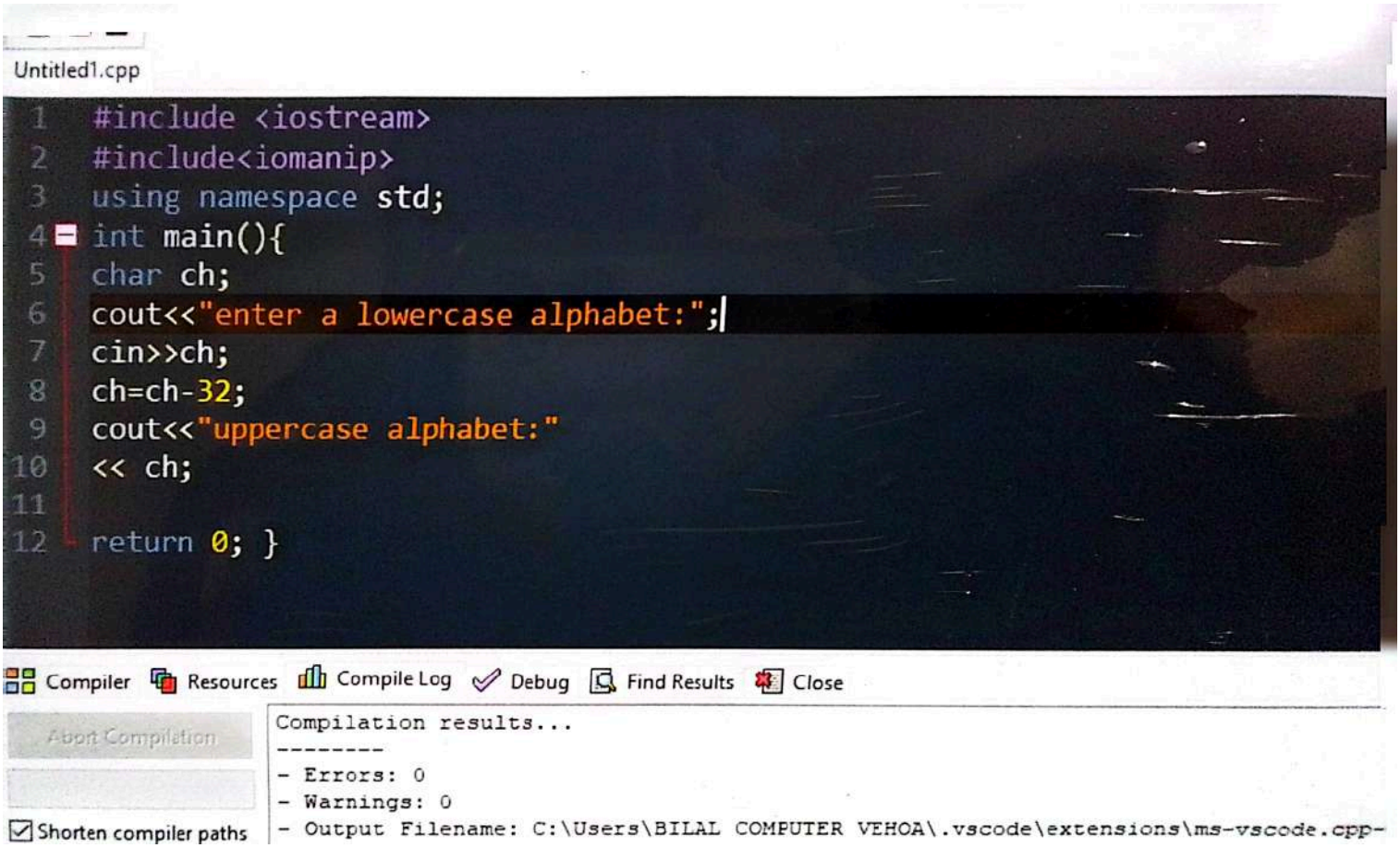
```
1   #include <iostream>
2   #include<iomanip>
3   Using namespace std;
4     int main(){
5     char ch;
6   cout<<"enter a lowercase alphabet:";
7     cin>>ch;
8     ch=ch-32;
9     cout<<"uppercase alphabet:"
10     <<ch;
11     return 0;}
```

Output:-

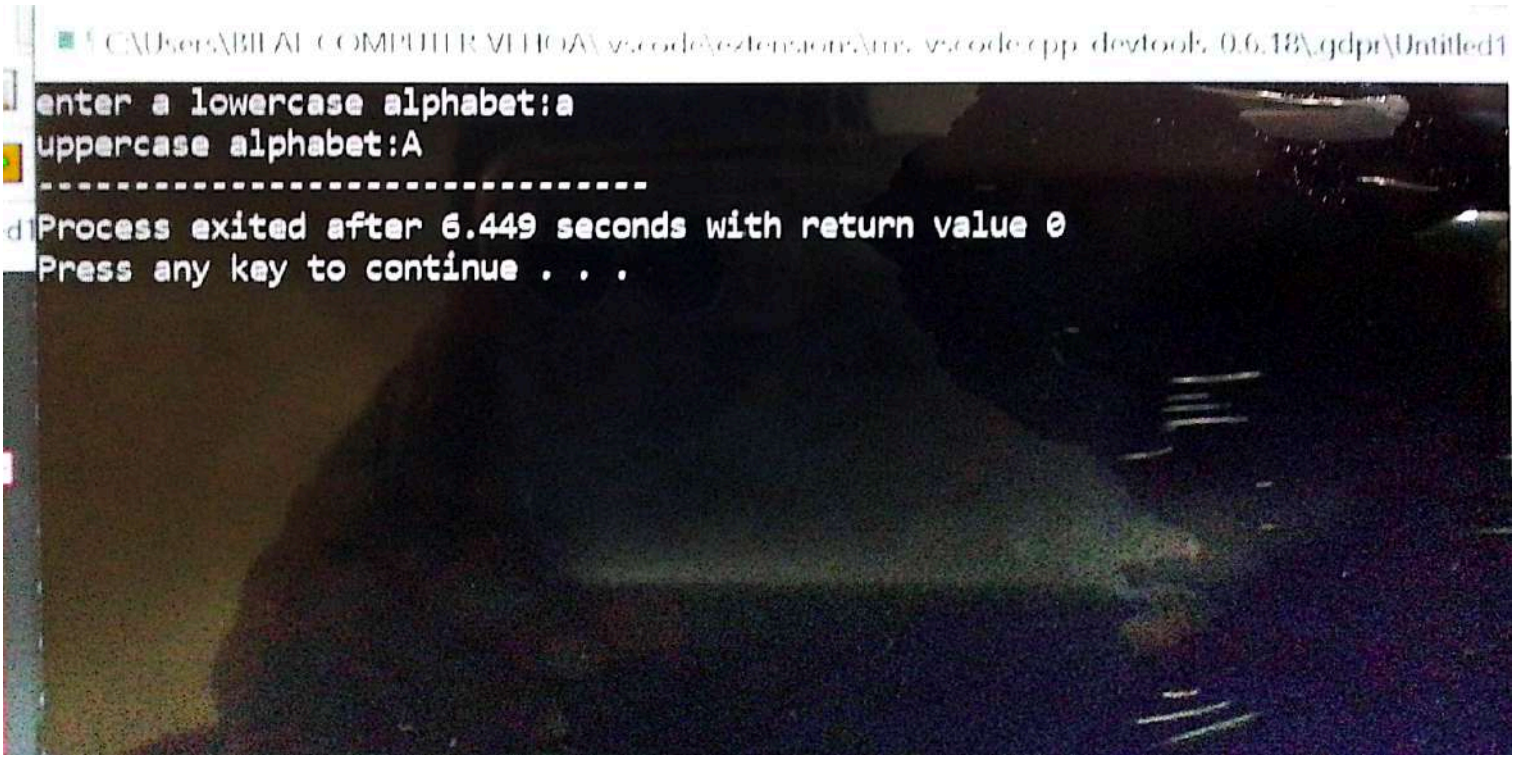
```
enter a lowercase alphabet:a
  uppercase alphabet:A
```

End:




Untitled1.cpp
#include <iostream>
#include<iomanip>
using namespace std;
int main(){
char ch;
cout<<"enter a lowercase alphabet:";
cin>>ch;
ch=ch-32;
cout<<"uppercase alphabet:"
<< ch;
return 0; }
Compiler
Resources
Compile Log
Debug
Find Results
Close
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpp-


C:\Users\BILAL COMPUTER VEHOA\.vscode\extensions\ms-vscode.cpptools-0.6.18\gdpr\Untitled1
enter a lowercase alphabet:a
uppercase alphabet:A

Process exited after 6.449 seconds with return value 0
Press any key to continue . . .